How Do Sectoral Shocks Shape Future GDP Growth?

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Predicting GDP growth relevant for many (monetary, fiscal, industrial) policy questions

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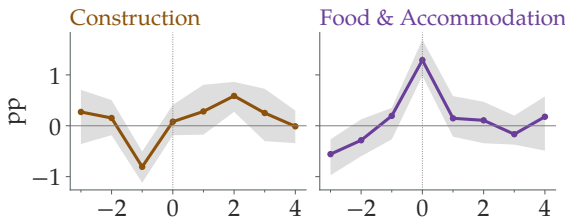
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Same Domar weight, about 12% of GDP: GDP growth on the sector's TFP innovation



Three Sectors

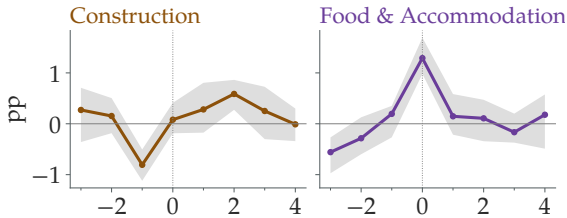
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Question: what explains the differences between the two sectors in these GDP dynamics?

Three Sectors

This Paper

Dynamic Hulten theorem

- Decompose Domar weights into impact and propagation components
- General production and accumulation; surprise and anticipated shocks

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GDP responses along any transitional path

- Characterize GDP dynamics through investment and hours
- Connect future GDP to the propagation component

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Quantification: eleven U.S. sectors

- Capture differences in GDP dynamics across sectors
- Production structure explains sectoral propagation rankings
- Rankings robust to preferences, accumulation, adjustment costs, and utilization

Prior Work

Aggregation in networks: Domar weights summarize first-order effects

Domar (1961); Hulten (1978); Gabaix (2011); Acemoglu et al. (2012); Baqaee and Farhi (2019); Yeh (2025)
Boehm et al. (2019); Carvalho et al. (2021); Miranda-Pinto et al. (2023); Silva et al. (2024)

Productivity and welfare: the present value of a productivity gain

Hulten (1979); Basu and Fernald (2002); Basu et al. (2022); Adamovsky et al. (2024)

Investment networks: investment amplifies and redirects shocks

Horvath (1998); Dupor (1999); Foerster et al. (2011); Atalay (2017); vom Lehn and Winberry (2022)
Foerster et al. (2022); Coşar et al. (2024); Casal and Caunedo (2024); Baqaee and Malmberg (2025)

Dynamics in networks: frictions and lags shape the timing of responses

Long and Plosser (1983); Kleinman et al. (2023); Afrouzi and Bhattarai (2023); Minton and Wheaton (2023)
Liu and Tsyvinski (2024); Leng et al. (2025); Liu et al. (2025); Schaal and Taschereau-Dumouchel (2025)

News and anticipation: expected productivity moves activity before it changes

Beaudry and Portier (2006, 2007); Jaimovich and Rebelo (2009)
Barsky and Sims (2011); Schmitt-Grohé and Uribe (2012)

Outline

Theory

Data

Quantitative Exercise

Conclusion

Environment

Production: N sectors, each with a constant-returns technology $\mathcal{F}_i$

$$Q_{it} = Z_{it}\, \mathcal{F}_i\left(L_{it},\, u_{it}K_{it},\, \boldsymbol{M}_{it}\right)$$

- Q_{it}: gross output; Z_{it}: productivity; $\boldsymbol{M}_{it}$: intermediate inputs
- u_{it}: utilization rate for the capital in place
- Labor is mobile across sectors

Capital Transactions Extension: Input Relationships

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Accumulation: each sector builds its own capital, with constant returns in $\mathcal{K}_i$

$$K_{i,t+1} = \mathcal{K}_i\big(K_{it},\, \boldsymbol{X}_{it};\, u_{it}\big)$$

- $\boldsymbol{X}_{it}$: investment goods bought from all sectors
- High utilization of capital at t potentially leaves less for $t+1$

Capital Transactions | Extension: Input Relationships

Real GDP

Real changes: quantities valued at reference prices

$$\mathrm{d}\log Y_t \equiv \sum_i S_{it}\,\mathrm{d}\log Y_{it} \qquad S_{it} \equiv \widetilde{Y}_{it}/\widetilde{Y}_t$$

- $\mathrm{d}\log Y_{it}$: change in sector i's real value added
- Tildes mark nominal values: $\widetilde{Y}_{it}$ is sector i's nominal value added

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Expenditure: GDP is consumed or invested

$$\mathrm{d}\log Y_t = S_{Ct}\,\mathrm{d}\log C_t + S_{Xt}\,\mathrm{d}\log X_t$$

- Consumption shares, by good: $S_{Ci,t} \equiv \widetilde{C}_{it}/\widetilde{Y}_t$; $S_{Ct} = \sum_i S_{Ci,t}$
- Investment shares, by buying sector: $S_{Xi,t} \equiv \widetilde{X}_{it}/\widetilde{Y}_t$; $S_{Xt} = \sum_i S_{Xi,t}$
- Labor share, wage W_t: $S_{Lt} \equiv W_t L_t/\widetilde{Y}_t$

Investment and Capital Value

Investment: three changes in the value of capital

$$S_{Xt}\,\mathrm{d}\log X_t = \underbrace{S_{Kt}\,\mathrm{d}\log K_t^s}_{\text{capital services}} + \underbrace{S_{Kt}^{+}\,\mathrm{d}\log K_{t+1}}_{\text{capital carried forward}} - \underbrace{\frac{1+r_{t-1}}{1+g_{t-1}}\,S_{K,t-1}^{+}\,\mathrm{d}\log K_t}_{\text{capital inherited}}$$

- Capital-income shares, rental rate R_{it}: $S_{Ki,t} \equiv R_{it}K_{it}/\widetilde{Y}_t$; $S_{Kt} = \sum_i S_{Ki,t}$
- Asset-to-income ratios, capital price $P_{i,t+1}^K$: $S_{Ki,t}^{+} \equiv P_{i,t+1}^K K_{i,t+1}/\widetilde{Y}_t$; $S_{Kt}^{+} = \sum_i S_{Ki,t}^{+}$

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- r_t: real interest rate measured using the GDP deflator
- g_t: real GDP growth

Real Changes in Full | Rates and Discounting

Investment and Capital Value

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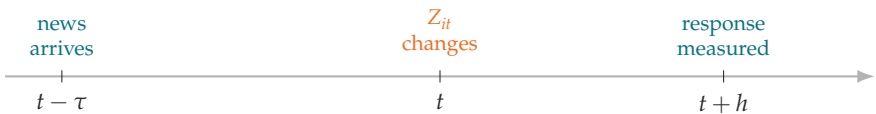
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- r_t: real interest rate measured using the GDP deflator
- g_t: real GDP growth
- When capital in place is fixed, the last term drops out

Real Changes in Full | Rates and Discounting

A Sectoral Shock, Possibly Anticipated

Experiment: a one-time change in sector i's productivity



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- A surprise if $\tau = 0$; anticipated if $\tau > 0$
- After t, productivity is back on its reference path

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Experiment: a one-time change in sector i's productivity



- A surprise if $\tau = 0$; anticipated if $\tau > 0$
- After t, productivity is back on its reference path
- First-order responses at every date $t+h$, $h \geq -\tau$
- Around a perfect-foresight reference path

Growth Accounting Along the Response

(Static) Hulten Theorem (1978): efficient production within each period

$$\frac{\mathrm{d}\log Y_{t+h}}{\mathrm{d}\log Z_{it}} = D_{it}\,\mathbf{1}\{h=0\} + S_{K,t+h}\frac{\mathrm{d}\log K^s_{t+h}}{\mathrm{d}\log Z_{it}} + S_{L,t+h}\frac{\mathrm{d}\log L_{t+h}}{\mathrm{d}\log Z_{it}}, \qquad h \geq -\tau$$

- Domar weight: $D_{it} = P_{it}Q_{it}/\widetilde{Y}_t$, gross output relative to GDP

Extension: Markups

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- Domar weight: $D_{it} = P_{it}Q_{it}/\widetilde{Y}_t$, gross output relative to GDP
- Away from the shock date, GDP moves only via capital services and hours
- Reallocation across sectors adds no separate term at first order

Extension: Markups

Hulten's Theorem with Capital Accumulation

Efficient production and accumulation imply, for every $T \geq 0$

$$D_{it} = \sum_{t'=t-\tau}^{t+T} \beta^g_{t,t'} \underbrace{\left(S_{C,t'} \frac{\mathrm{d} \log C_{t'}}{\mathrm{d} \log Z_{it}} - S_{L,t'} \frac{\mathrm{d} \log L_{t'}}{\mathrm{d} \log Z_{it}} \right)}_{\text{consumption net of hours}} + \beta^g_{t,t+T} \underbrace{S^+_{K,t+T} \frac{\mathrm{d} \log K_{t+T+1}}{\mathrm{d} \log Z_{it}}}_{\text{capital carried forward}}$$

Assumptions Discounting

Extension: Markups Extension: Adjustment Costs Extension: Open Economy Extension: Input Relationships

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- Growth-adjusted discounting: $\beta^{g}_{t,t+h}$ compounds $(1+g)/(1+r)$, a net rate $\approx r - g$
- Production efficiency alone — no household optimality

Assumptions | Discounting
Extension: Markups | Extension: Adjustment Costs | Extension: Open Economy | Extension: Input Relationships

Impact and Propagation Components

The Domar weight splits as $D_{it} = D^c_{it} + D^x_{it}$

$$\underbrace{D^c_{it} \equiv S_{Ct} \frac{\mathrm{d}\log C_t}{\mathrm{d}\log Z_{it}} - S_{Lt} \frac{\mathrm{d}\log L_t}{\mathrm{d}\log Z_{it}}}_{\text{impact component}} \qquad \underbrace{D^x_{it} \equiv S_{Xt} \frac{\mathrm{d}\log X_t}{\mathrm{d}\log Z_{it}} - S_{Kt} \frac{\mathrm{d}\log K^s_t}{\mathrm{d}\log Z_{it}}}_{\text{propagation component}}$$

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- D_{it}^{c} can be negative: extra hours may pay for investment, not consumption
- D_{it}^{x} can be negative: shock-date consumption net of hours may exceed the windfall

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By the investment identity: capital carried forward less capital inherited

$$D^x_{it} = S^+_{Kt} \frac{\mathrm{d}\log K_{t+1}}{\mathrm{d}\log Z_{it}} - \frac{1 + r_{t-1}}{1 + g_{t-1}} S^+_{K,t-1} \frac{\mathrm{d}\log K_t}{\mathrm{d}\log Z_{it}}$$

- Fixed utilization: after a surprise: $D^x_{it} = S_{Xt}\, \mathrm{d}\log X_t / \mathrm{d}\log Z_{it}$, impact investment

Propagation of a Surprise Shock

After a surprise, $\tau = 0$, for every $T \geq 0$

$$D^x_{it} = \sum_{t<t' \leq t+T} \beta^g_{t,t'} \left(S_{C,t'} \frac{\mathrm{d} \log C_{t'}}{\mathrm{d} \log Z_{it}} - S_{L,t'} \frac{\mathrm{d} \log L_{t'}}{\mathrm{d} \log Z_{it}} \right) + \beta^g_{t,t+T} S^+_{K,t+T} \frac{\mathrm{d} \log K_{t+T+1}}{\mathrm{d} \log Z_{it}}$$

All Later Dates

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- At $T = 0$: the value of the capital the shock leaves behind
- If terminal capital vanishes: the present value of later consumption net of hours

All Later Dates

Propagation of an Anticipated Shock

With news at $t-\tau$, $\tau>0$, for every $T\geq 0$

$$D^x_{it}=\sum_{t<t'\leq t+T}\beta^g_{t,t'}\left(S_{C,t'}\frac{\mathrm{d}\log C_{t'}}{\mathrm{d}\log Z_{it}}-S_{L,t'}\frac{\mathrm{d}\log L_{t'}}{\mathrm{d}\log Z_{it}}\right)+\beta^g_{t,t+T}S^+_{K,t+T}\frac{\mathrm{d}\log K_{t+T+1}}{\mathrm{d}\log Z_{it}}$$
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Statement and Financing | Pre-Shock Windows

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- Same form as after a surprise, net of capital accumulated before t

Statement and Financing Pre-Shock Windows

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- Same form as after a surprise, net of capital accumulated before t
- Financing: capital inherited at t is forgone pre-shock consumption net of hours

Statement and Financing | Pre-Shock Windows

Household Preferences

Standard: time separable, over a CRS consumption aggregate $C = \mathcal{C}(\boldsymbol{C})$

$$\mathcal{V}_{t-\tau} = \sum_{t' \geq t-\tau} \beta^{t'-t+\tau} U(C_{t'}, L_{t'})$$

- $\mathcal{V}_{t-\tau}$: the continuation objective from the news date, given the past
- Log-separable: $U(C, L) = \log C - \mathcal{G}(L)$, with $\mathcal{G}$ the disutility of hours
- GHH: $U(C, L) = \log[C - \mathcal{G}(L)]$, no wealth effect on hours

  Greenwood et al. (1988)

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General: the household ranks whole paths of consumption and hours

$$\mathcal{V}_{t-\tau} \equiv \mathcal{V}\left(\{\boldsymbol{C}_{t+h}, L_{t+h}\}_{h \geq -\tau}\right)$$

- Need not be time separable (used in the quantitative model)

  Jaimovich and Rebelo (2009)

General Preferences

Domar Weights and Welfare

Time-separable preferences and optimal household choices imply

Hulten (1979); Basu and Fernald (2002); Basu et al. (2022); Baqaee and Rubbo (2023)

$$D_{it} = S_{Ct} \sum_{t' \geq t-\tau} \beta^{t'-t} \frac{\mathrm{d}U_{t'} / \mathrm{d} \log Z_{it}}{\partial U_t / \partial \log C_t}$$

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Impact component: the welfare contribution of the shock date

$$D_{it}^{c} = S_{Ct} \frac{\mathrm{d}U_t / \mathrm{d} \log Z_{it}}{\partial U_t / \partial \log C_t}$$

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$$D^c_{it} = S_{Ct} \frac{\mathrm{d}U_t / \mathrm{d} \log Z_{it}}{\partial U_t / \partial \log C_t}$$

Propagation component: the welfare contribution of all other dates since the news

$$D^x_{it} = S_{Ct} \left(\sum_{t'=t-\tau}^{t-1} \beta^{t'-t} \frac{\mathrm{d}U_{t'} / \mathrm{d} \log Z_{it}}{\partial U_t / \partial \log C_t} + \sum_{t'>t} \beta^{t'-t} \frac{\mathrm{d}U_{t'} / \mathrm{d} \log Z_{it}}{\partial U_t / \partial \log C_t} \right)$$

Assumptions General Preferences

Response of GDP to Surprise/Anticipated Sectoral TFP Shock

Along an equilibrium path, with capital the only state and fixed utilization, for $h \geq -\tau$

$$\frac{\mathrm{d}\log Y_{t+h}}{\mathrm{d}\log Z_{it}} = D_{it}\mathbf{1}\{h=0\} + \left(S'_{K,t+h} + S_{L,t+h}\frac{\partial \log L_{t+h}}{\partial \log \boldsymbol{K}_{t+h}}\right)\frac{\mathrm{d}\log \boldsymbol{K}_{t+h}}{\mathrm{d}\log Z_{it}} + S_{L,t+h}\frac{\partial \log L_{t+h}}{\partial \log Z_{it}}\mathbf{1}\{h \leq 0\}$$

where

$$\frac{\mathrm{d}\log \boldsymbol{K}_{t+h}}{\mathrm{d}\log Z_{it}} = \sum_{t'=t-\tau}^{\min(t,t+h-1)} \mathcal{J}_{t+h-1}\cdots\mathcal{J}_{t'+1}\,\zeta_{d,t'}\,\frac{\partial \log \boldsymbol{X}_{t'}}{\partial \log Z_{it}}$$

$$\mathcal{J}_t \equiv \boldsymbol{I} - \zeta_{d,t} + \zeta_{d,t}\frac{\partial \log \boldsymbol{X}_t}{\partial \log \boldsymbol{K}_t} \qquad \zeta_{j,t} \equiv \frac{S_{Xj,t}}{S^{+}_{Kj,t}} \text{ in the diagonal } \zeta_{d,t}$$

- Shock-specific, only from $t-\tau$ to t: D_{it} and the responses at given capital $\partial/\partial \log Z_{it}$
- Common to all shocks: the policies $\partial/\partial \log \boldsymbol{K}$ of investment $\boldsymbol{X}$ and hours; $\boldsymbol{S}_K$, S_L, $\boldsymbol{\zeta}_d$

Surprise Case | News and States | GDP at Every Date

One Period After a Surprise

The GDP path at $h = 1$: next-period GDP through impact investment alone

$$\frac{\mathrm{d}\log Y_{t+1}}{\mathrm{d}\log Z_{it}} = \Big(\underbrace{S'_{X,t}\,\iota_{d,t}}_{\text{capital income}} + \underbrace{S_{L,t+1}\frac{\partial \log L_{t+1}}{\partial \log \boldsymbol{K}_{t+1}}\,\zeta_{d,t}}_{\text{hours}} \Big) \frac{\mathrm{d}\log \boldsymbol{X}_t}{\mathrm{d}\log Z_{it}}$$

Yield: next-period capital income per unit of capital value, each relative to its date's GDP

$$\iota_{jt} \equiv \frac{S_{Kj,t+1}}{S^{+}_{Kj,t}} \quad \text{in the diagonal } \boldsymbol{\iota}_{d,t}$$

Yields

One Period After a Surprise

The GDP path at $h = 1$: next-period GDP through impact investment alone

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Equal yields, $\iota_{jt} = \iota_t$: the propagation component returns at the yield

$$\frac{\mathrm{d}\log Y_{t+1}}{\mathrm{d}\log Z_{it}} = \iota_t\, D^{x}_{it} + S_{L,t+1}\frac{\partial \log L_{t+1}}{\partial \log \boldsymbol{K}_{t+1}}\,\boldsymbol{\zeta}_{d,t}\,\frac{\mathrm{d}\log \boldsymbol{X}_t}{\mathrm{d}\log Z_{it}}$$

Yields

One Period After a Surprise

The GDP path at $h = 1$: next-period GDP through impact investment alone

$$\frac{\mathrm{d}\log Y_{t+1}}{\mathrm{d}\log Z_{it}} = \Big(\underbrace{\boldsymbol{S}'_{X,t}\,\boldsymbol{\iota}_{d,t}}_{\text{capital income}} + \underbrace{S_{L,t+1}\frac{\partial \log L_{t+1}}{\partial \log \boldsymbol{K}_{t+1}}\,\boldsymbol{\zeta}_{d,t}}_{\text{hours}} \Big) \frac{\mathrm{d}\log \boldsymbol{X}_t}{\mathrm{d}\log Z_{it}}$$

Yield: next-period capital income per unit of capital value, each relative to its date's GDP

$$\iota_{jt} \equiv \frac{S_{Kj,t+1}}{S^{+}_{Kj,t}} \quad \text{in the diagonal } \boldsymbol{\iota}_{d,t}$$

Equal yields, $\iota_{jt} = \iota_t$: the propagation component returns at the yield

$$\frac{\mathrm{d}\log Y_{t+1}}{\mathrm{d}\log Z_{it}} = \iota_t\, D^{x}_{it} + S_{L,t+1}\frac{\partial \log L_{t+1}}{\partial \log \boldsymbol{K}_{t+1}}\,\boldsymbol{\zeta}_{d,t}\,\frac{\mathrm{d}\log \boldsymbol{X}_t}{\mathrm{d}\log Z_{it}}$$

- With unequal yields, where impact investment goes changes the capital contribution

Yields

What Moves Hours?

Log-separable: $\mathcal{G}'(L)\,L = S_L / S_C$

- Hours move only if the labor share moves relative to the consumption share
- Common Cobb–Douglas capital share: hours rise with the investment share S_X

Log-Separable | No Wealth Effect | Amplification | Before the Shock

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GHH: $\mathcal{G}'(L) = W/P^C$, with P^C the consumption deflator

- No wealth effect: hours rise when the wage buys more consumption
- Benchmark: impact hours rise only with gains that reach consumer goods
  - Geometric, GHH, per unit of D_i: Construction 0.11, Trade 0.77, Food & Acc. 0.90

Log-Separable | No Wealth Effect | Amplification | Before the Shock

Production and Aggregation

In every specification: Cobb–Douglas production and investment bundles

$$Q_{it} = Z_{it}\left(K_{it}^{\alpha_i} L_{it}^{1-\alpha_i}\right)^{\gamma_i} \prod_j M_{jit}^{(1-\gamma_i)\phi_{ji}} \qquad X_{it} = \prod_j \left(\frac{X_{jit}}{\omega_{ji}}\right)^{\omega_{ji}}$$

- γ_i: value-added share of gross output; α_i: capital share of value added
- Materials $\mathbf{\Phi}$: ϕ_{ji} is good j's share in sector i's materials
- Investment $\mathbf{\Omega}$: ω_{ji} is good j's share in sector i's investment

Notation

Production and Aggregation

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Aggregation: the Leontief inverse and the Domar weights, as shares of GDP

$$\boldsymbol{\mathcal{L}} \equiv \underbrace{\left(\boldsymbol{I} - (\boldsymbol{I} - \boldsymbol{\gamma}_d)\boldsymbol{\Phi}'\right)^{-1}}_{\text{all rounds of input use}} \qquad \boldsymbol{D} = \boldsymbol{\mathcal{L}}' \underbrace{\left(\boldsymbol{S}_C + \boldsymbol{\Omega}\,\boldsymbol{S}_X\right)}_{\text{final spending on each good}}$$

Notation

Accumulation and Preferences

Capital accumulation

- Geometric: $K_{i,t+1} = K_{it}^{1-\zeta_i} X_{it}^{\zeta_i}$
- Linear: $K_{i,t+1} = (1-\delta_i)K_{it} + X_{it}$

Preferences

- Log-separable: $\log C - \chi L$, so that $\chi L = S_L / S_C$: hours move with spending shares
- GHH: $\log\left[C - \chi L^{1+\psi}/(1+\psi)\right]$ ($\psi = 1$, unit Frisch elasticity); no wealth effect

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Three economies: their GDP paths differ only through investment and hours responses

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- Geometric accumulation, log-separable preferences: hours constant

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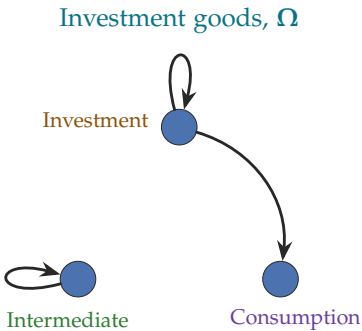
Three economies: their GDP paths differ only through investment and hours responses

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- Geometric accumulation, GHH preferences
- Linear accumulation, log-separable preferences

vom Lehn and Winberry (2022)

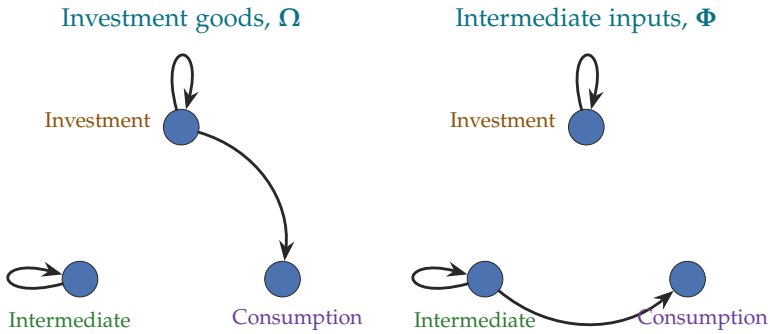
Three Sectors, Three Roles

Two networks: arrows run from supplier to buyer



Three Sectors, Three Roles

Two networks: arrows run from supplier to buyer

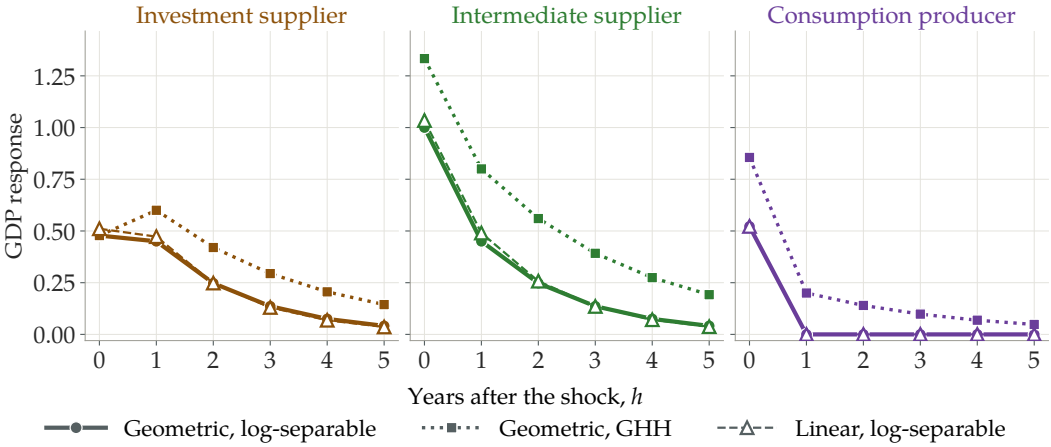


- Akin to Construction, Trade (Wholesale & Retail), and Food & Accommodation

Limiting Example

Propagation in the Benchmark

GDP responses in the three-sector economy, $\zeta_i = \delta_i = 0.9$ for all sectors



GHH Policies Parameters

Data

Annual U.S. industry data: eleven private sectors, excluding agriculture and government

- Networks: BEA input–output tables; capital-flow tables for investment goods
  vom Lehn and Winberry (2022)
- TFP: gross-output TFP from the BEA–BLS KLEMS accounts
- Years: networks averaged over 1948–2024; regressions on 1951–2020

Definitions

The Three Sectors and the Stylized Economy

Stylized counterparts: investment supplier, intermediate supplier, consumption producer

	U.S. economy				Stylized economy	
	Supplies, % of all investment	Supplies, % of all intermediates	D_i	D_i^x/D_i	D_i	D_i^x/D_i
Construction	36.9	1.4	0.12	0.89	1/2	1
Trade	6.8	13.8	0.29	0.21	1	1/2
Food & Accommodation	0.2	3.7	0.12	0.08	1/2	0

- D_i^x/D_i: the Geometric, log-separable economy; for the stylized one, limits $\beta, \zeta \to 1$

The Three Sectors and the Stylized Economy

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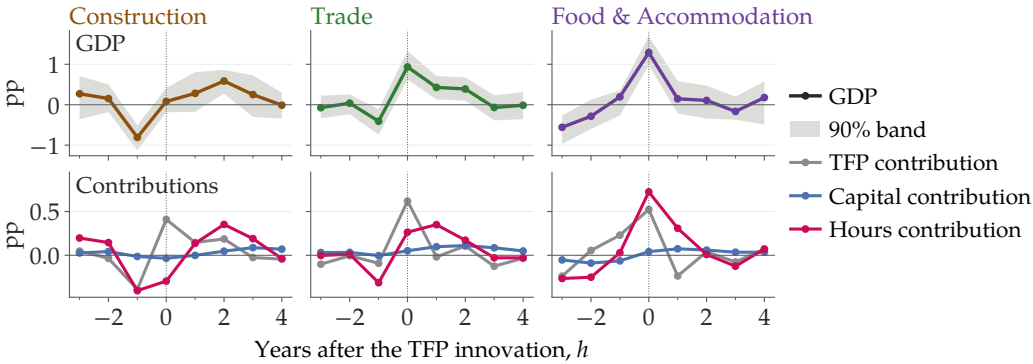
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- D_i^x/D_i: the Geometric, log-separable economy; for the stylized one, limits $\beta, \zeta \to 1$
- Construction and Food & Accommodation: similar D_i, opposite D_i^x/D_i
- Trade: the largest Domar weight of the three, but 50% of its sales go to consumers

U.S. Networks | Three-Sector Example

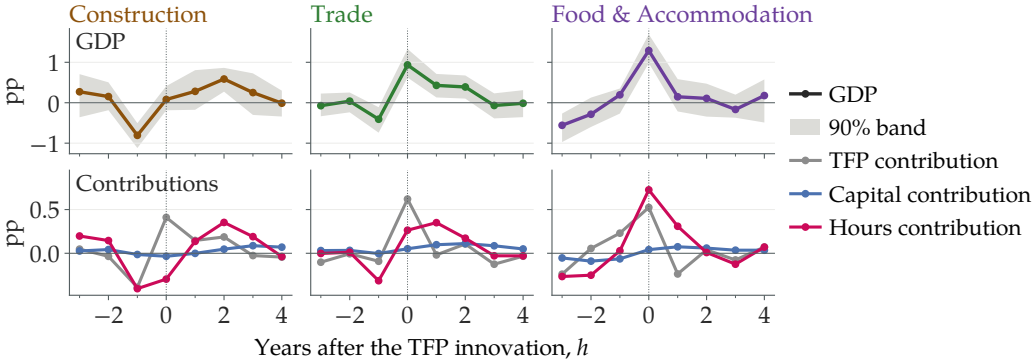
Three Sectors in the Data

Regression coefficients: GDP growth and its growth-accounting contributions



Three Sectors in the Data

Regression coefficients: GDP growth and its growth-accounting contributions



- After impact, the hours contribution is usually the largest
- The capital contribution is small throughout

The Quantitative Model

Capital accumulation: Ψ_i, costs of investment growth; $\delta_i(u)$, rising with utilization

Christiano, Eichenbaum and Evans (2005)

$$K_{i,t+1} = \left[1 - \Psi_i(X_{it}/X_{i,t-1})\right] X_{it} + \left(1 - \delta_i(u_{it})\right) K_{it}$$

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Preferences: Jaimovich–Rebelo; index $\overline{C}_t$, with weight κ on current consumption

Jaimovich and Rebelo (2009)

$$E_0 \sum_t \beta^t \log\left(C_t - \chi \overline{C}_t \frac{L_t^{1+\psi}}{1+\psi}\right) \qquad \overline{C}_t = C_t^{\kappa} \overline{C}_{t-1}^{1-\kappa}$$

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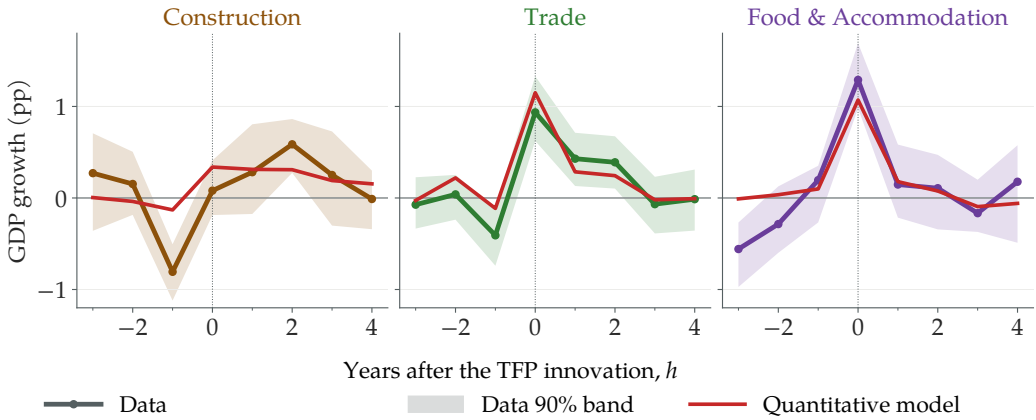
TFP process: surprise $\widetilde{\varepsilon}_0$; trend news a year ahead, $\widetilde{\nu}_1$, and on arrival, $\widetilde{\nu}_0$

$$\Delta z_{it} = \rho_i\, \Delta z_{i,t-1} + (1-\rho_i)\, g^z_{i,t-1} + \widetilde{\varepsilon}_{0,it} \qquad g^z_{it} = \rho^g g^z_{i,t-1} + \widetilde{\nu}_{1,it} + \widetilde{\nu}_{0,i,t+1}$$

- $z_{it} = \log Z_{it}$; ρ_i, ρ^g: persistence; a share s of each trend innovation is news
- Surprises share one common factor, with a signed loading per sector

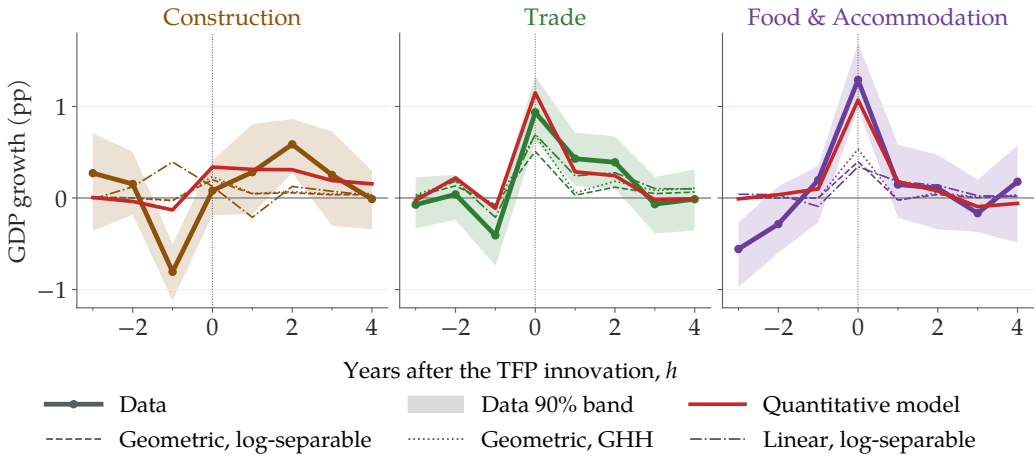
The Quantitative Model and the Data

GDP growth on a sector's TFP innovation: the data and the quantitative model



The Quantitative Model and the Data

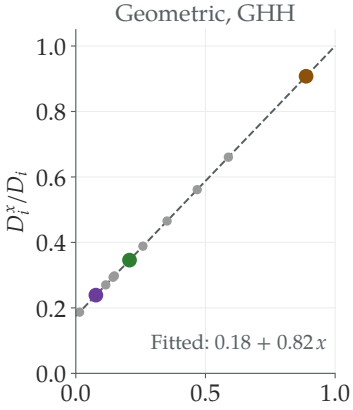
GDP growth on a sector's TFP innovation: the data and the quantitative model



GDP and Its Contributions | Fit by Block | Hours and Consumption | All Sectors | Correlations | From the Benchmark | Real Estate | News

Propagation Shares across Economies

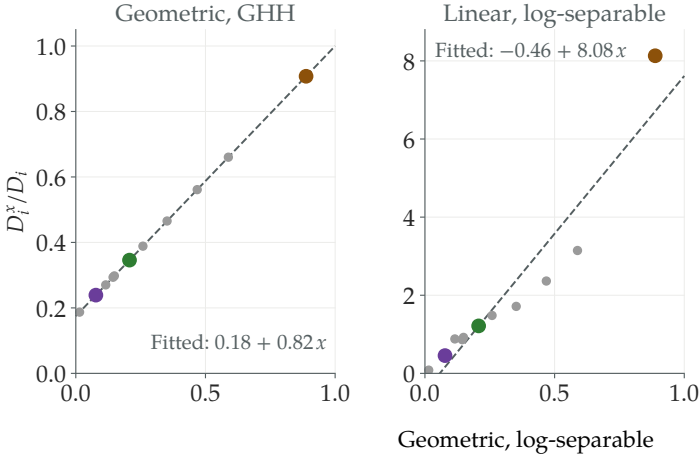
Propagation share D_i^x / D_i, against the geometric, log-separable economy



Geometric, log-separable

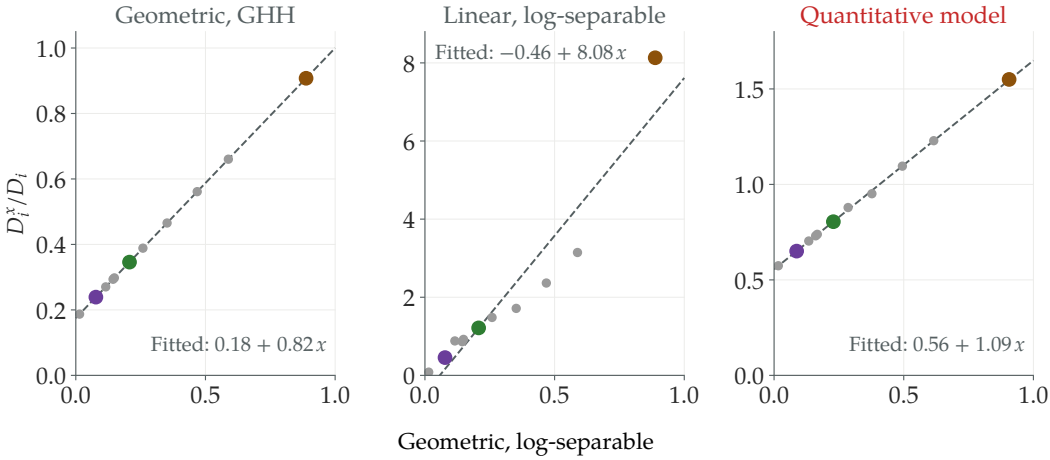
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Propagation Shares across Economies

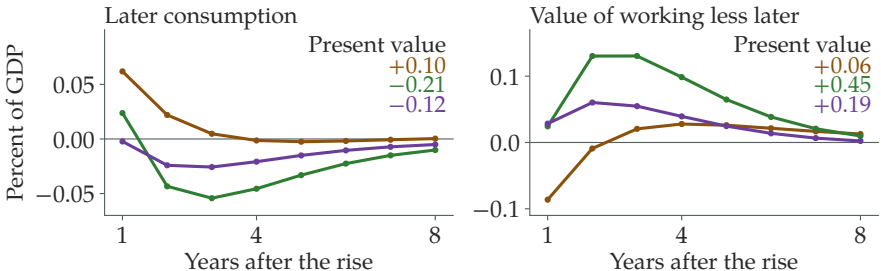
Propagation share D_i^x / D_i, against the geometric, log-separable economy



What the Propagation Component Delivers

Quantitative model, one-date 1% rise: Construction, Trade, Food & Accommodation

$$D^x_{it} = \underbrace{\sum_{t'>t} \beta^g_{t,t'} S_{C,t'} \frac{\mathrm{d}\log C_{t'}}{\mathrm{d}\log Z_{it}}}_{\text{later consumption}} + \underbrace{\left(-\sum_{t'>t} \beta^g_{t,t'} S_{L,t'} \frac{\mathrm{d}\log L_{t'}}{\mathrm{d}\log Z_{it}}\right)}_{\text{value of working less later}}$$

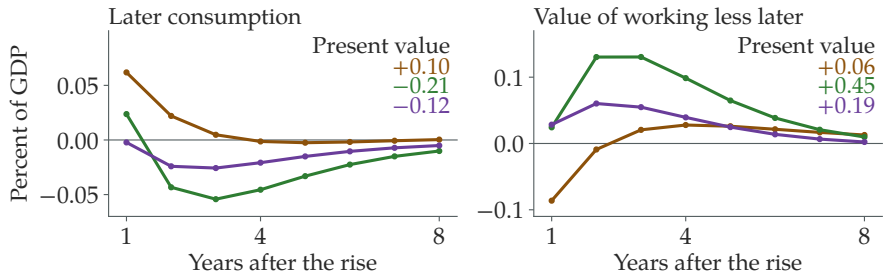


Capital Carried Forward

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Construction: the gain returns as later consumption

Trade, Food & Accommodation: it returns as working less

Capital Carried Forward

Conclusion

Theory: Hulten's theorem with capital accumulation, from production efficiency alone

- Domar weights fix a present value, split into impact and propagation, $D_{it} = D^c_{it} + D^x_{it}$
- Investment and hours shape the GDP path

Data: Construction, Food & Accommodation: similar Domar weights, opposite paths

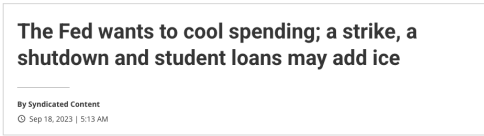
- After impact, the hours contribution is usually the largest

Quantitative: propagation shares rank sectors almost identically in all four economies

- Construction: the gain returns as later consumption
- Trade, Food & Accommodation: it returns as working less

Extensions

A Sectoral Shock and the GDP Forecast



Reuters, September 2023: a strike as a risk to the outlook



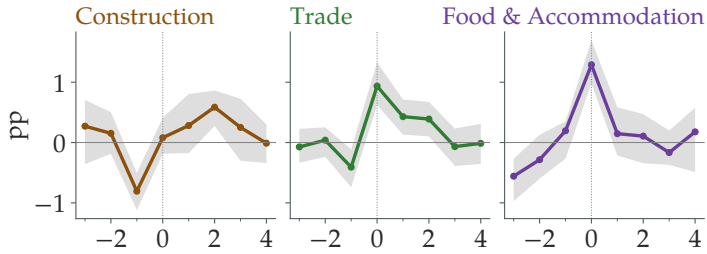
FEDS Note: Q4 GDP growth -0.5 pp

What happens after the initial GDP loss?

Back

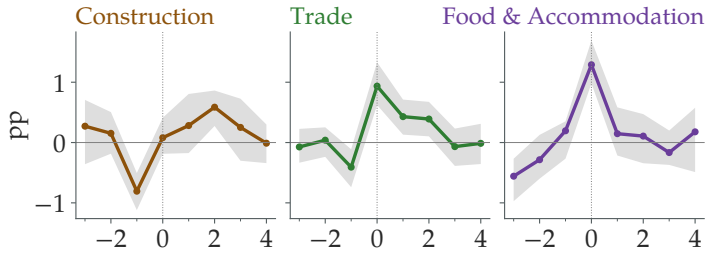
Three Sectors, Three Dynamics

Regression coefficients: GDP growth associated with sectoral TFP innovations



Three Sectors, Three Dynamics

Regression coefficients: GDP growth associated with sectoral TFP innovations



- Construction: small on impact, peaks two years later
- Trade: large impact, persists for about two years
- Food & Accommodation: large impact, rapid fade

Specification | All Sectors | Local Projections | Back

The Opening Figure: Specification

Regression: GDP growth at $t+h$ on sector i's TFP innovation at t

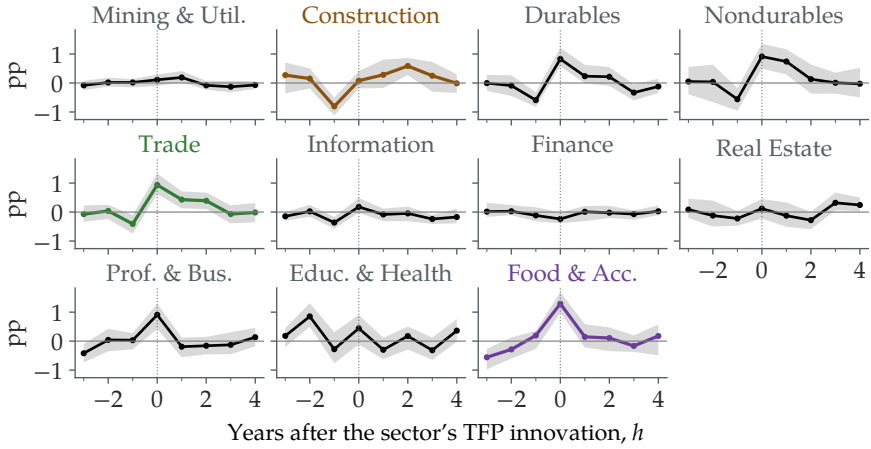
$$b_{i,h} \equiv \frac{\text{Cov}\left(\Delta \log Y_{t+h},\, \varepsilon_{it}\right)}{\text{Var}(\varepsilon_{it})}, \qquad h = -3, \ldots, 4$$

- One regression for each lead or lag h, with no other controls
- ε_{it}: TFP innovation, the residual of an AR(1) in sector i's TFP growth
- Y: U.S. private non-farm value added, eleven KLEMS sectors
- Years $t = 1951, \ldots, 2020$; 90% moving-block bootstrap bands
- h counts from the year the change shows up in TFP, not when it was learned

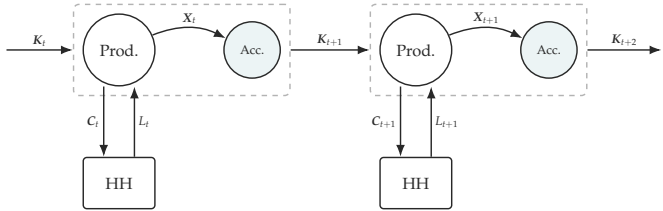
Back

All Eleven Sectors

Regression coefficients: GDP growth at $t + h$ on a sector's TFP innovation at t



Capital Transactions and Prices



- Producers buy capital at P^K_{it} and sell it back at $P^K_{it} - R_{it}$
- Capital producers combine it with investment and sell $K_{i,t+1}$ at $P^K_{i,t+1}$
- Arbitrage: $P^K_{i,t+1} = (1 + \widetilde{r}_t)\, P^K_{i,t+1}$, with $\widetilde{r}_t$ the nominal interest rate
- Nominal GDP: $\widetilde{Y}_t = \widetilde{C}_t + \widetilde{X}_t \equiv \sum_i P_{it} C_{it} + \sum_{j,i} P_{jt} X_{jit}$

◂ Environment ◂ Real GDP

Real Changes in Full

Share-weighted sums of sectoral log changes, at reference prices

$$\mathrm{d}\log Y_{it} \equiv \frac{P_{it}\,\mathrm{d}Q_{it} - \sum_j P_{jt}\,\mathrm{d}M_{jit}}{\widetilde{Y}_{it}}, \qquad \mathrm{d}\log K^s_t \equiv \sum_i \frac{R_{it}K_{it}}{\widetilde{K}^s_t}\left(\mathrm{d}\log u_{it} + \mathrm{d}\log K_{it}\right),$$

$$\mathrm{d}\log C_t \equiv \sum_i \frac{P_{it}C_{it}}{\widetilde{C}_t}\,\mathrm{d}\log C_{it}, \qquad \mathrm{d}\log K_t \equiv \sum_i \frac{P^K_{it}K_{it}}{\widetilde{K}_t}\,\mathrm{d}\log K_{it},$$

$$\mathrm{d}\log X_t \equiv \sum_{j,i} \frac{P_{jt}X_{jit}}{\widetilde{X}_t}\,\mathrm{d}\log X_{jit}, \qquad \mathrm{d}\log L_t \equiv \sum_i \frac{W_{it}L_{it}}{\widetilde{W}_t}\,\mathrm{d}\log L_{it}$$

- Capital services: $\widetilde{K}^s_t \equiv \sum_i R_{it}K_{it}$
- Capital stock: $\widetilde{K}_t \equiv \sum_i P^K_{it}K_{it}$
- Labor compensation: $\widetilde{W}_t \equiv \sum_i W_{it}L_{it}$
- With a common wage, $\mathrm{d}\log L_t$ is the change in aggregate hours

◂ Real GDP ◂ Capital Value

Growth Accounting with Markups

Solow residual: $\mathrm{d}\log \overline{Z}_{t+h} \equiv \mathrm{d}\log Y_{t+h} - S_{L,t+h}\,\mathrm{d}\log L_{t+h} - S_{K,t+h}\,\mathrm{d}\log K^{s}_{t+h}$

Without markups: Hulten's theorem

$$\frac{\mathrm{d}\log \overline{Z}_{t+h}}{\mathrm{d}\log Z_{it}} = D_{it}\,\mathbf{1}\{h=0\}$$

With markups: $\mu_{j,t+h} \geq 1$, unchanged by the shock

$$\frac{\mathrm{d}\log \overline{Z}_{t+h}}{\mathrm{d}\log Z_{it}} = \frac{D_{it}}{\mu_{it}}\,\mathbf{1}\{h=0\} + \sum_{j} \pi_{j,t+h}\frac{\mathrm{d}\log Q_{j,t+h}}{\mathrm{d}\log Z_{it}}$$

- $\pi_{j,t+h} \equiv (1 - 1/\mu_{j,t+h})\,D_{j,t+h}$: sector j's profits relative to GDP
- Output responses can move measured productivity before and after the shock
- CRS and cost minimization; factor payments exclude profits

  Hall (1988); Basu and Fernald (2002); Baqaee and Farhi (2020); Osotimehin and Popov (2023)

Capital Accumulation ◂ Extensions ◂ Back

Hulten's Theorem with Capital Accumulation and Markups

Extension: markups, for every $T \geq -\tau$

$$\sum_{t'=t-\tau}^{t+T} \beta_{t,t'}^{g} \frac{\mathrm{d}\log \overline{Z}_{t'}}{\mathrm{d}\log Z_{it}} = \sum_{t'=t-\tau}^{t+T} \beta_{t,t'}^{g} \left(S_{C,t'} \frac{\mathrm{d}\log C_{t'}}{\mathrm{d}\log Z_{it}} - S_{L,t'} \frac{\mathrm{d}\log L_{t'}}{\mathrm{d}\log Z_{it}} \right) + \beta_{t,t+T}^{g} S_{K,t+T}^{+} \frac{\mathrm{d}\log K_{t+T+1}}{\mathrm{d}\log Z_{it}}$$

- Without markups, the left side is D_{it} for $T \geq 0$, as on the main slide
- With markups, it is the present value of the residual's responses

  Basu and Fernald (2002); Basu et al. (2022)
- Competitive accumulation and asset markets; fixed utilization
- Capital is the only inherited stock, unchanged at $t - \tau$

Growth Accounting ◂ Extensions ◂ Capital Accumulation

Capital Accumulation: Assumptions

Only production efficiency and inherited capital

- Efficient production within each period, with supporting prices
- Efficient accumulation of capital, with supporting asset prices
- Capital is the only inherited production stock
- Capital in place at the news date $t - \tau$ is unchanged

No assumption on household preferences or optimality

Back

Growth-Adjusted Discounting

Rates

- GDP deflator: $P_t^Y \equiv \widetilde{Y}_t / Y_t$
- Real interest rate: $1 + r_t \equiv (1 + \widetilde{r}_t)\, P_t^Y / P_{t+1}^Y$
- Real GDP growth: $1 + g_t \equiv Y_{t+1} / Y_t$

Discount factor:

$$\beta^g_{t,t+h} \equiv \prod_{t''=t}^{t+h-1} \frac{1+g_{t''}}{1+r_{t''}} \quad (h \geq 0) \qquad\qquad \beta^g_{t+h,t} \equiv 1/\beta^g_{t,t+h}$$

- One-period net rate $(r_t - g_t)/(1+g_t)$, approximately $r_t - g_t$
- Depends only on the nominal interest rate and nominal GDP

◂ Capital Value ◂ Capital Accumulation

The Propagation Component over All Later Dates

After a surprise, as $T \to \infty$: if the discounted sum converges and

$$\lim_{T\to\infty} \beta^g_{t,t+T}\, S^+_{K,t+T} \frac{\mathrm{d}\log K_{t+T+1}}{\mathrm{d}\log Z_{it}} = 0,$$

$$D^x_{it} = \sum_{t'>t} \beta^g_{t,t'} \left(S_{C,t'} \frac{\mathrm{d}\log C_{t'}}{\mathrm{d}\log Z_{it}} - S_{L,t'} \frac{\mathrm{d}\log L_{t'}}{\mathrm{d}\log Z_{it}} \right)$$

- The propagation component is the present value of later consumption net of hours
- Under the same assumptions

Back

News and the Financing Identity

As stated in the paper: for every $T \geq 0$

$$D_{it}^{x} = \sum_{\substack{t-\tau \leq t' \leq t+T \\ t' \neq t}} \beta_{t,t'}^{g} \left(S_{C,t'} \frac{\mathrm{d} \log C_{t'}}{\mathrm{d} \log Z_{it}} - S_{L,t'} \frac{\mathrm{d} \log L_{t'}}{\mathrm{d} \log Z_{it}} \right) + \beta_{t,t+T}^{g} S_{K,t+T}^{+} \frac{\mathrm{d} \log K_{t+T+1}}{\mathrm{d} \log Z_{it}}$$

Financing identity: pre-shock consumption net of hours, compounded to t

$$\sum_{t'=t-\tau}^{t-1} \beta_{t,t'}^{g} \left(S_{C,t'} \frac{\mathrm{d} \log C_{t'}}{\mathrm{d} \log Z_{it}} - S_{L,t'} \frac{\mathrm{d} \log L_{t'}}{\mathrm{d} \log Z_{it}} \right) = - \frac{1 + r_{t-1}}{1 + g_{t-1}} S_{K,t-1}^{+} \frac{\mathrm{d} \log K_{t}}{\mathrm{d} \log Z_{it}}$$

- Substituting the identity gives the main slide's form
- An accounting identity: no additional restriction

Components | Anticipated Shock

Windows Ending before the Shock

For every T with $-\tau \leq T < 0$:

$$0 = \sum_{t'=t-\tau}^{t+T} \beta^{g}_{t,t'} \left(S_{C,t'} \frac{\mathrm{d} \log C_{t'}}{\mathrm{d} \log Z_{it}} - S_{L,t'} \frac{\mathrm{d} \log L_{t'}}{\mathrm{d} \log Z_{it}} \right) + \beta^{g}_{t,t+T} S^{+}_{K,t+T} \frac{\mathrm{d} \log K_{t+T+1}}{\mathrm{d} \log Z_{it}}$$

- Before the shock, responses are movements along the frontier
- The window ending at $t-1$ gives the financing identity
- Under the same assumptions

Back

Welfare: Assumptions

Added to the production assumptions

- Time-separable preferences over a CRS consumption aggregate
- Productivity does not enter preferences directly
- Consumption and intertemporal conditions hold at market prices
- The hours condition holds unless hours are fixed
- The discounted utility sum can be differentiated term by term
- Discounted flow-utility responses are summable
- The discounted terminal capital response vanishes

Welfare

General Preferences

Reading: D_{it}/S_{Ct} is the welfare gain as an equivalent change in $\log C_t$

- Under log-separable utility, the normalization $\partial U_t/\partial \log C_t$ equals one

General $\mathcal{V}_{t-\tau}$, including habit: the same statements hold

- Each date's term values the quantities dated $t+h$

With habit: D^c_{it} need not equal the response of date-t flow utility

Assumptions ◂ Preferences ◂ Welfare

General Preferences: Assumptions

With general preferences $\mathcal{V}_{t-\tau}$

- Household conditions hold at market prices
- Productivity does not enter preferences directly
- Any habit is internalized
- Initial preference states are fixed
- The welfare derivative equals the convergent sum of dated contributions
- The discounted terminal capital response vanishes

Back

The GDP Response: Investment Form

Investment form: at every date $t+h$, $h \geq -\tau$

$$\frac{\mathrm{d}\log Y_{t+h}}{\mathrm{d}\log Z_{it}} = D_{it}\mathbf{1}\{h=0\} + S_{L,t+h}\frac{\mathrm{d}\log L_{t+h}}{\mathrm{d}\log Z_{it}} + \boldsymbol{S}'_{K,t+h}\sum_{t'=t-\tau}^{t+h-1}\left[\prod_{t''=t'+1}^{t+h-1}\left(\boldsymbol{I}-\boldsymbol{\zeta}_{d,t''}\right)\right]\boldsymbol{\zeta}_{d,t'}\frac{\mathrm{d}\log \boldsymbol{X}_{t'}}{\mathrm{d}\log Z_{it}}$$

- Each sector's capital is the decayed sum of its own investment since the news
- $\zeta_{j,t+h} = S_{Xj,t+h}/S^{+}_{Kj,t+h}$: investment relative to the capital value it builds
- Assumes efficient production and accumulation, and fixed utilization

Saving Form ◂ Back

The GDP Response: Saving Form

Saving form: with equal yields at $t+h-1$, for $h > -\tau$

$$\frac{\mathrm{d}\log Y_{t+h}}{\mathrm{d}\log Z_{it}} = D_{it}\mathbf{1}\{h=0\} + S_{L,t+h}\frac{\mathrm{d}\log L_{t+h}}{\mathrm{d}\log Z_{it}} + \frac{\iota_{t+h-1}}{\beta^g_{t,t+h-1}}\Big[D^x_{it}\mathbf{1}\{h\geq 1\} - \sum_{\substack{t-\tau\leq t'\leq t+h-1\\ t'\neq t}} \beta^g_{t,t'}\Big(S_{C,t'}\frac{\mathrm{d}\log C_{t'}}{\mathrm{d}\log Z_{it}} - S_{L,t'}\frac{\mathrm{d}\log L_{t'}}{\mathrm{d}\log Z_{it}}\Big)\Big]$$

- The bracket is the part of the windfall still saved at $t+h-1$
- Assumes efficient production and accumulation, and fixed utilization

Back

Rental Yields and Durability

Yield: $\iota_{jt} \equiv S_{Kj,t+1}/S^{+}_{Kj,t}$, shares of GDP at their own dates

Yield identity, with efficient production and accumulation:

$$\iota_{jt} = \frac{1+r_t}{1+g_t} - (1-\zeta_{j,t+1})\frac{S^{+}_{Kj,t+1}}{S^{+}_{Kj,t}}$$

- With linear accumulation, longer-lived capital has a lower gross yield
- Holding interest rates, growth and capital gains fixed
- Next-period capital income: fixed utilization and a surprise, $\tau = 0$, suffice

Back

The GDP Response after a Surprise

After a surprise, $\tau = 0$, for every $h \geq 1$

$$\frac{\mathrm{d}\log Y_{t+h}}{\mathrm{d}\log Z_{it}} = \underbrace{\left(S'_{K,t+h} + S_{L,t+h}\frac{\partial \log L_{t+h}}{\partial \log \boldsymbol{K}_{t+h}}\right)}_{\text{GDP response to capital}} \underbrace{\mathcal{J}_{t+h-1}\cdots\mathcal{J}_{t+1}}_{\text{transitions}}\zeta_{d,t} \underbrace{\frac{\mathrm{d}\log \boldsymbol{X}_t}{\mathrm{d}\log Z_{it}}}_{\text{impact investment}}$$

- On impact, the responses at given capital are the whole responses
- Capital is the only inherited state of a locally unique, differentiable equilibrium
- At given capital: re-solved from $t+h$, current capital at its reference value
- Same impact investment response $\Rightarrow$ same GDP response from $t+1$ on

Prices Back

Cobb–Douglas Prices

Cobb–Douglas production eliminates prices

$$\begin{aligned}\mathrm{d}\log X_{j,t+h} = &\Big[\boldsymbol{\Omega}'\boldsymbol{\mathcal{L}}\Big(\boldsymbol{e}_i\,\mathrm{d}\log Z_{it}\,\mathbf{1}\{h=0\} + \boldsymbol{\gamma}_d\boldsymbol{\alpha}_d\big(\mathrm{d}\log \boldsymbol{K}_{t+h} + \mathrm{d}\log \boldsymbol{u}_{t+h}\big)\\ &\qquad + \boldsymbol{\gamma}_d(\boldsymbol{I}-\boldsymbol{\alpha}_d)\mathbf{1}\,\mathrm{d}\log L_{t+h}\Big)\Big]_j\\ &+ \mathrm{d}\log S_{Xj,t+h} - \Big[\boldsymbol{\Omega}'\boldsymbol{\mathcal{L}}\boldsymbol{\gamma}_d\big(\boldsymbol{\alpha}_d\,\mathrm{d}\log \boldsymbol{D}_{t+h} + (\boldsymbol{I}-\boldsymbol{\alpha}_d)\mathbf{1}\,\mathrm{d}\log S_{L,t+h}\big)\Big]_j\end{aligned}$$

- Efficient within-period production at supporting prices
- One mobile labor input
- Coefficients evaluated at $t+h$, $h \geq -\tau$
- Spending and hours policies are still needed to close investment

Notation ◂ Equilibrium ◂ Back

Cobb–Douglas Notation

Production and expenditure weights

- γ_j: value-added share of gross output
- α_j: capital share of value added
- ϕ_{ij}: good i's share in sector j's materials, collected in $\mathbf{\Phi}$
- $\omega_{ij,t+h}$: good i's share in buyer j's investment, collected in $\mathbf{\Omega}$
- $\boldsymbol{\mathcal{L}} \equiv [\boldsymbol{I} - (\boldsymbol{I} - \boldsymbol{\gamma}_d)\mathbf{\Phi}']^{-1}$: all rounds of intermediate-input use

◂ Prices ◂ Benchmark

News and Additional States

With news: investment at given capital, from $t - \tau$ to t, builds capital at $t + 1$

- For $\tau > 0$, at the news date capital has not moved: GDP responds only through hours
- Stationary reference: the responses depend only on the periods left until t

Additional inherited states enlarge the transition

- Adjustment costs: lagged investment; impact investment still sets both states
- Habit or the JR index: the impact consumption response is needed too
- Variable utilization: a within-date policy, with no new state

Adjustment Costs ◂ Back

Hours under Log-Separable Preferences

Interior labor supply: $\mathcal{G}'(L)\,L = S_L/S_C$, so that

$$(1+\psi)\,\mathrm{d}\log L = \mathrm{d}\log S_L - \mathrm{d}\log S_C, \qquad \psi \equiv L\mathcal{G}''/\mathcal{G}', \text{ the inverse Frisch elasticity}$$

- Common Cobb–Douglas capital share: $(1+\psi)\,\mathrm{d}\log L = (S_X/S_C)\,\mathrm{d}\log S_X$
- Investment reallocated across buyers at the same total share leaves hours unchanged

Geometric accumulation, common α and ζ, Cobb–Douglas value added

- Interior efficient equilibrium; fixed utilization; household transversality
- $S_C = 1 - \beta\zeta\alpha/[1-\beta(1-\zeta)]$, so hours do not respond to TFP, at any date or lead
- Any substitution elasticities in the other CES nests

Back

Hours without a Wealth Effect

GHH, Cobb–Douglas production: $\boldsymbol{\theta} \equiv \boldsymbol{S}_C / S_C$, current consumption weights, at $t + h$

$$(\psi + \boldsymbol{\theta}' \boldsymbol{\mathcal{L}} \boldsymbol{\gamma}_d \boldsymbol{\alpha}_d \mathbf{1}) \frac{\mathrm{d} \log L}{\mathrm{d} \log Z_{it}} = \boldsymbol{\theta}' \boldsymbol{\mathcal{L}} \left[\boldsymbol{e}_i \mathbf{1}\{h = 0\} + \boldsymbol{\gamma}_d \boldsymbol{\alpha}_d \frac{\mathrm{d} \log \boldsymbol{K}}{\mathrm{d} \log Z_{it}} - \boldsymbol{\gamma}_d \boldsymbol{\alpha}_d \frac{\mathrm{d} \log \boldsymbol{D} - \mathbf{1}\, \mathrm{d} \log S_L}{\mathrm{d} \log Z_{it}} \right]$$

- $\boldsymbol{\theta}' \boldsymbol{\mathcal{L}} \boldsymbol{\gamma}_d \boldsymbol{\alpha}_d \mathbf{1}$: capital's cost share in the consumption basket; $\boldsymbol{e}_i$: the shocked sector
- Positive left coefficient: productivity and capital raise hours at fixed share responses
- Last term: capital-income shares against the labor share in consumer-goods supply
- Before the shock at given capital, only the restarted share-response term remains

Benchmark Back

Hours without a Wealth Effect: Benchmark

Constant spending shares: the last term vanishes, and $\mathcal{G} = \chi L^{1+\psi}/(1+\psi)$

$$\frac{\mathrm{d}\log L_{t+h}}{\mathrm{d}\log Z_{it}} = \frac{\boldsymbol{\theta}'\boldsymbol{\mathcal{L}}\left(e_i\mathbf{1}\{h=0\} + \boldsymbol{\gamma}_d\boldsymbol{\alpha}_d\,\mathrm{d}\log \boldsymbol{K}_{t+h}/\mathrm{d}\log Z_{it}\right)}{1+\psi-\boldsymbol{\theta}'\boldsymbol{\mathcal{L}}\boldsymbol{\gamma}_d(\boldsymbol{I}-\boldsymbol{\alpha}_d)\mathbf{1}}$$

- On impact, gains reaching consumer goods move hours; later, their suppliers' capital
- Same positive one-date gain: GHH GDP is at least as large at every horizon

Three-sector economy: $\alpha = \gamma = \frac{1}{2}$, $\beta = 0.96$, $\zeta = 0.9$, unit Frisch elasticity

- Investment supplier: no impact hours; GDP response 0.48 on impact, 0.60 next period
- Consumption producer: hours rise on impact, so investment and later GDP rise too

GHH Policies ◂ Back

Hours from Inherited Capital

GHH: one-date TFP change, fixed utilization, capital the only state
From the date after the change, even if it was announced earlier
$1/\psi$: Frisch elasticity; relative price: GDP deflator / consumption deflator

$$\text{hours' GDP contribution} = \frac{S_L}{1+\psi-S_L} \times \text{capital's GDP contribution}$$

- Condition: capital's labor-share and relative-price log derivatives sum to zero
- $S_L = 0.58332$: multiplier 0.41 at unit Frisch; upper limit 1.40
- Actual Geometric, GHH: ratios 0.36–0.43 in years 1–2; condition not imposed

Spending shares can break this link

- Linear, log-separable: Construction, year 1: hours -0.26 pp; capital -0.02 pp

Back

Hours before the Shock

Time-separable, capital-only equilibrium; strict saving curvature: at given capital, $h < 0$

$$\frac{\partial \log L_{t+h}}{\partial \log Z_{it}} = \left.\frac{\partial \log L_{t+h}}{\partial \log \boldsymbol{K}_{t+h+1}}\right|_{\boldsymbol{K}_{t+h},\boldsymbol{Z}_{t+h}} \zeta_{d,t+h} \frac{\partial \log \boldsymbol{X}_{t+h}}{\partial \log Z_{it}}$$

- Hours add no shock-specific information beyond pre-shock investment
- Pre-shock investment and hours vanish iff the gain's capital gradient is zero
- Benchmark, either preference: no real allocations respond before the shock

Log-separable, power labor cost, common Cobb–Douglas capital share, one basket

$$\frac{\partial \log L_{t+h}}{\partial \log Z_{it}} = \frac{S'_{X,t+h}\, \partial \log \boldsymbol{X}_{t+h} / \partial \log Z_{it}}{(1-\alpha) + (\psi+\alpha) S_{C,t+h}}$$

Two Sectors ◂ Back

Anticipation without Hours: Two Sectors

Two sectors: log-separable, power labor cost, no intermediate inputs

- Common Cobb–Douglas capital share; common geometric elasticity $0 < \zeta < 1$
- Same CES basket for consumption and both buyers, elasticity $\sigma^C > 0$; stationary path

A gain in sector i announced one period ahead

- $\sigma^C > 1$: buyer i invests more and the other buyer less; signs reverse if $\sigma^C < 1$
- Hours do not move: constant hours do not rule out investment anticipation

Back

The Benchmark Response

Geometric, log-separable: spending shares and hours constant; for $h \geq 1$

$$\frac{\mathrm{d}\log Y_{t+h}}{\mathrm{d}\log Z_{it}} = S_K' \mathcal{J}^{h-1} \zeta_d \underbrace{\Omega' \mathcal{L} e_i}_{\text{impact investment}}$$

where

$$\mathcal{J} = I - \zeta_d + \zeta_d \Omega' \mathcal{L} \gamma_d \alpha_d$$

- $\Omega' \mathcal{L} e_i$: the gain from sector i that reaches each buyer's investment goods
- $\mathcal{J}$: persistence of each stock plus investment out of capital

Related closed forms: Long and Plosser (1983); Horvath (1998); Dupor (1999)
Foerster et al. (2011); vom Lehn and Winberry (2022)

Back

Benchmark: Primitives

The benchmark economy

- Cobb–Douglas production, with fixed $\boldsymbol{\alpha}_d$, γ_d, $\boldsymbol{\Phi}$ and positive labor shares
- Cobb–Douglas investment bundles $\boldsymbol{\Omega}$ and consumption aggregate $\boldsymbol{\theta}$
- Geometric accumulation, with $0 < \zeta_j \leq 1$; fixed utilization
- Log-separable, or GHH with $\mathcal{G}(L) = \chi L^{1+\psi}/(1+\psi)$ and $\chi, \psi > 0$
- Positive initial capital; an interior and finite optimum

Hence: constant spending shares, $\beta^g_{t,t+h} = \beta^h$, and yields $\iota_j = 1/\beta - 1 + \zeta_j$

Notation Back

Benchmark Policies

Log-separable: hours row zero, $\partial \log \boldsymbol{X} / \partial \log \boldsymbol{K} = \boldsymbol{\Omega}' \boldsymbol{\mathcal{L}} \boldsymbol{\gamma}_d \boldsymbol{\alpha}_d$; one period later

$$\frac{\mathrm{d} \log Y_{t+1}}{\mathrm{d} \log Z_{it}} = \sum_j \alpha_j \gamma_j D_j \zeta_j [\boldsymbol{\Omega}' \boldsymbol{\mathcal{L}}]_{ji}$$

GHH: $\mathcal{G}(L) = \chi L^{1+\psi} / (1+\psi)$

$$\frac{\partial \log L}{\partial \log \boldsymbol{K}} = \frac{\boldsymbol{\theta}' \boldsymbol{\mathcal{L}} \boldsymbol{\gamma}_d \boldsymbol{\alpha}_d}{1 + \psi - \boldsymbol{\theta}' \boldsymbol{\mathcal{L}} \boldsymbol{\gamma}_d (\boldsymbol{I} - \boldsymbol{\alpha}_d) \mathbf{1}} \qquad \frac{\partial \log \boldsymbol{X}}{\partial \log \boldsymbol{K}} = \boldsymbol{\Omega}' \boldsymbol{\mathcal{L}} \boldsymbol{\gamma}_d \left(\boldsymbol{\alpha}_d + (\boldsymbol{I} - \boldsymbol{\alpha}_d) \mathbf{1} \frac{\partial \log L}{\partial \log \boldsymbol{K}} \right)$$

- Hours respond to the gain that reaches the consumption basket

◂ Benchmark ◂ Figure ◂ Hours

Benchmark Properties

No anticipation: no real allocation moves before t, for any lead

Cumulative response, under log-separable preferences:

$$\sum_{h \geq 0} \frac{\mathrm{d} \log Y_{t+h}}{\mathrm{d} \log Z_{it}} = \boldsymbol{D}' \left(\boldsymbol{I} - \gamma_d \boldsymbol{\alpha}_d \boldsymbol{\Omega}' \boldsymbol{\mathcal{L}} \right)^{-1} \boldsymbol{e}_i$$

Cancellation: hours stay constant with common α and ζ

- Log-separable utility, Cobb–Douglas value added, any CES elsewhere
- Interior efficient equilibrium; fixed utilization; geometric accumulation
- The household's transversality condition holds

Back

The Three-Sector Example

Log-separable, $\alpha = \gamma = 1/2$, limits $\beta, \zeta \to 1$

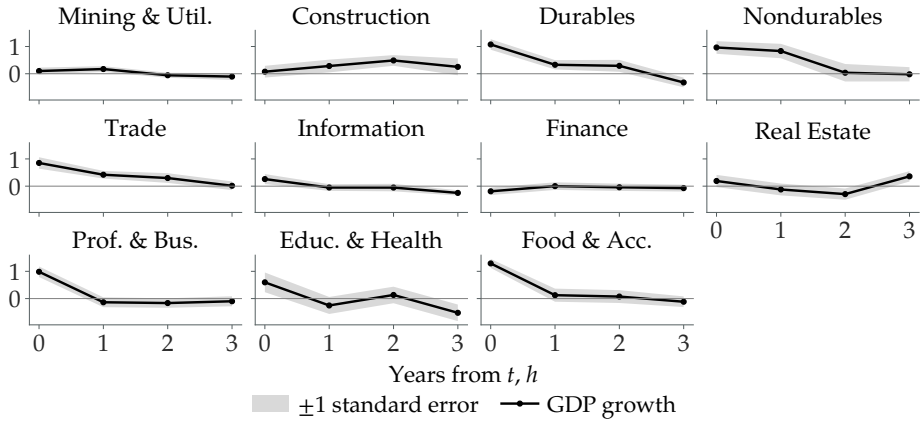
	D_i	D_i^c	D_i^x	GDP at $h = 1$
Investment supplier	1/2	0	1/2	1/2
Intermediate supplier	1	1/2	1/2	1/2
Consumption producer	1/2	1/2	0	0

- The figure uses $\beta = 0.96$, $\zeta_i = \delta_i = 0.9$ and a unit Frisch elasticity under GHH
- Linear, log-separable: $\delta_i = 0.9$, with no adjustment cost

◂ Roles ◂ Figure ◂ U.S. Sectors

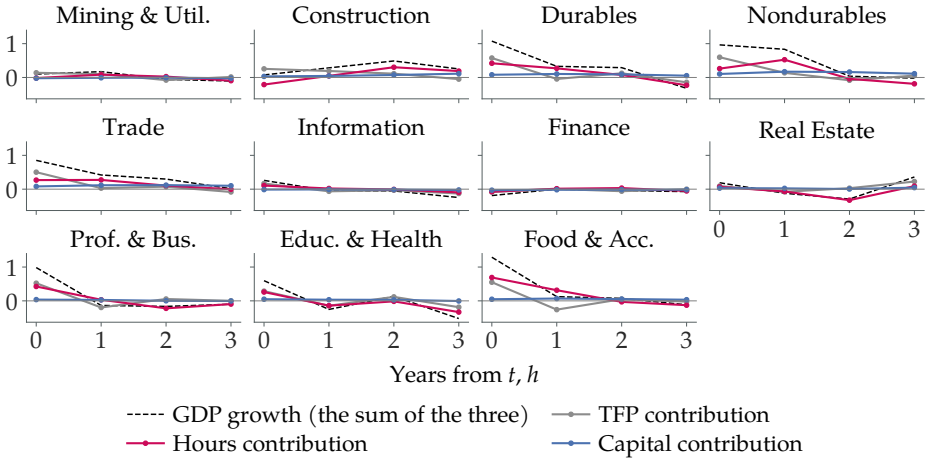
Local Projections: GDP

Specification: $h = 0, \ldots, 3$; two outcome and two TFP lags; ± 1 HAC s.e.



Contributions
Back

Local Projections: GDP and Its Contributions



GDP Growth before a Sector's TFP Rise

Local projections: one per sector and horizon

$$\Delta \log Y_{t+h} = \eta_{i,h} \Delta \log Z_{it} + \text{controls} + e_{i,t+h}$$

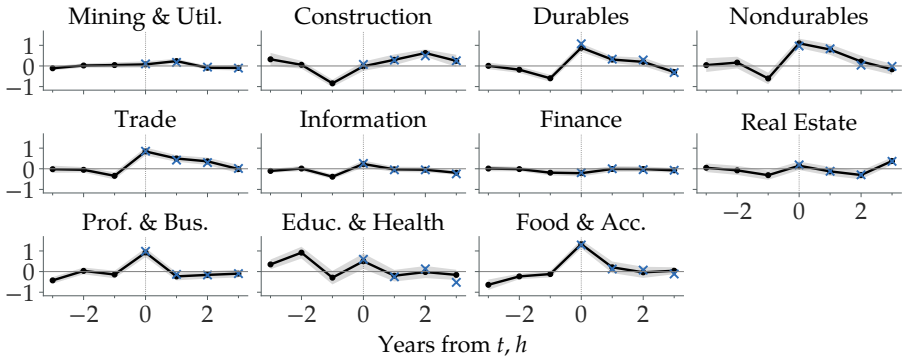
- Controls:
  - GDP growth at $t-4$ and $t-5$
  - The sector's TFP growth at $t-1$ and $t-2$
- $h = -3, \ldots, 3$; lines use $t = 1953, \ldots, 2021$
- Bands: ± 1 HAC standard error

The year before: negative in 10 of 11 sectors

- Largest for Construction: -0.84 (s.e. 0.18)
- A correlation, not evidence of anticipation

All Sectors' Coefficients ◂ All Sectors

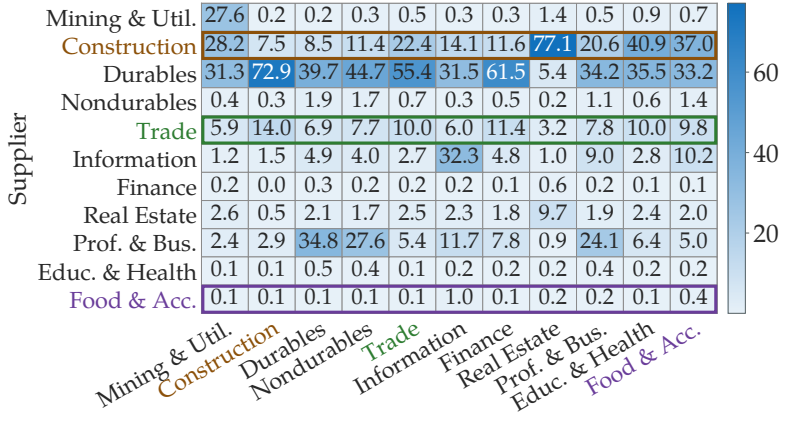
GDP Growth before a Sector's TFP Rise: All Sectors



Back

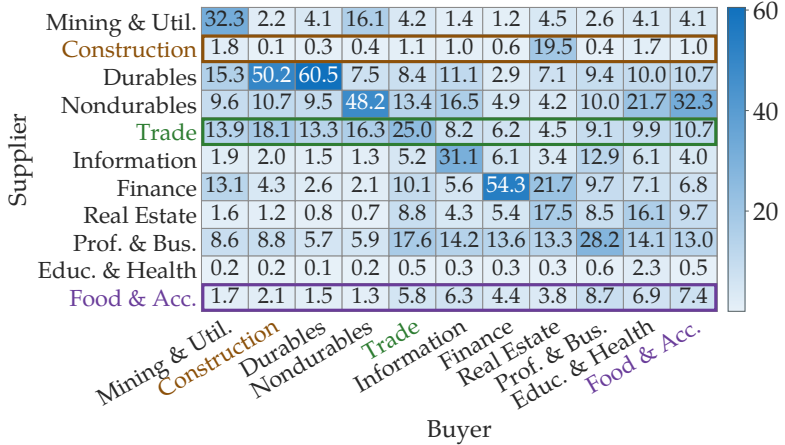
U.S. Input–Output Networks

Investment goods, Ω: each buyer's investment by supplier, in percent



U.S. Input–Output Networks

Intermediate inputs, Φ: each buyer's intermediate inputs by supplier, in percent



Data: Definitions and Uncertainty

Object: private non-farm output growth on the sector's TFP growth, 1951–2020

- TFP growth whitened with a sector AR(1)
- GDP split exactly into TFP, capital and hours contributions, at income shares

Uncertainty: nominal 90% bands cover 78–86% in simulation

- Coefficients correlate 0.94–0.99 with local projections controlling for lags

Food & Accommodation: KLEMS Other Services, excl. Government. Trade: Trade and Support Activities.

Back

Extensions

Extension	Relaxes	Main change
Adjustment costs	accumulation from K_t and $\boldsymbol{X}_t$	lagged investment is a state
Markups and wedges	efficient production	PDV of residual responses
Open economy	closed economy	terms of trade, foreign assets
Input relationships	capital the only stock	relationships carried forward

Adjustment Costs Markups Open Economy Input Relationships

Back

Hulten's Theorem with Adjustment Costs

Extension: lagged investment is a second inherited stock, for every $T \geq 0$

$$D_{it} = \sum_{t'=t-\tau}^{t+T} \beta^{g}_{t,t'} \left(S_{C,t'} \frac{\mathrm{d} \log C_{t'}}{\mathrm{d} \log Z_{it}} - S_{L,t'} \frac{\mathrm{d} \log L_{t'}}{\mathrm{d} \log Z_{it}} \right) + \beta^{g}_{t,t+T} \left(S^{+}_{K,t+T} \frac{\mathrm{d} \log K_{t+T+1}}{\mathrm{d} \log Z_{it}} + \sum_{j} S^{\Psi}_{j,t+T} \frac{\mathrm{d} \log X_{j,t+T}}{\mathrm{d} \log Z_{it}} \right)$$

- The terminal value adds lagged investment, at its value share $S^{\Psi}_{j,t+T}$
- $S^{\Psi}_{j,t+T}$ can be negative, and is zero when reference investment is constant
- For $-\tau \leq T < 0$, the same expression equals zero

Investment Adjustment Costs: GDP Response

Adjustment costs: fixed utilization

$$K_{j,t+h+1} = (1-\delta_j)K_{j,t+h} + \left[1 - \Psi_j(X_{j,t+h}/X_{j,t+h-1})\right]X_{j,t+h} \qquad \Psi_j(1) = \Psi_j'(1) = 0$$

- Lagged investment is a second inherited state, with value share $S^{\Psi}_{j,t+h}$
- Equal yields at $t+h-1$, $h > -\tau$: GDP includes

$$-\iota_{t+h-1} \sum_j S^{\Psi}_{j,t+h-1} \frac{\mathrm{d}\log X_{j,t+h-1}}{\mathrm{d}\log Z_{it}}$$

- At the stationary reference the correction vanishes

Assumptions | News and States | Extensions

Adjustment Costs: Assumptions

With investment adjustment costs

- Efficient production and accumulation at supporting prices
- Fixed utilization
- Capital and lagged investment are the only inherited production stocks
- Both inherited stocks are unchanged at $t - \tau$

Back

Hulten's Theorem in the Open Economy

Extension: open economy, with terms of trade $\mathcal{R}$ and foreign assets B^*, for $T \geq -\tau$

$$\sum_{t'=t-\tau}^{t+T} \beta^g_{t,t'} \left(\frac{\mathrm{d} \log \overline{Z}_{t'}}{\mathrm{d} \log Z_{it}} + \frac{\mathrm{d}\mathcal{R}_{t'}}{\mathrm{d} \log Z_{it}} \right)$$

$$= \sum_{t'=t-\tau}^{t+T} \beta^g_{t,t'} \left(S_{C,t'} \frac{\mathrm{d} \log C_{t'}}{\mathrm{d} \log Z_{it}} - S_{L,t'} \frac{\mathrm{d} \log L_{t'}}{\mathrm{d} \log Z_{it}} \right) + \beta^g_{t,t+T} \left(S^+_{K,t+T} \frac{\mathrm{d} \log K_{t+T+1}}{\mathrm{d} \log Z_{it}} + \frac{1}{\widetilde{Y}_{t+T}} \frac{\mathrm{d} B^*_{t+T+1}}{\mathrm{d} \log Z_{it}} \right)$$

- Efficient domestic production and accumulation at supporting purchaser prices
- Capital and foreign assets earn the world rate, which the shock does not move
- They are the only inherited stocks, both unchanged at $t - \tau$

◂ Extensions ◂ Capital Accumulation

Extension: Input Relationships

Extension: changing the use of an input is costly

Liu and Tsyvinski (2024)

$$M_{jit} = \mathcal{M}_{ji}\left(M_{ji,t-1},\, M^{p}_{jit}\right)$$

- M_{jit}: the input in use, which enters production
- M^{p}_{jit}: purchases of good j, used up within the period
- $\mathcal{M}_{ji}$: constant returns, increasing in purchases, concave
- The relationship used today is inherited tomorrow: state $(K_{it}, \boldsymbol{M}_{i,t-1})$
- Relationship prices: P^{M}_{jit} at the start of t, P^{M}_{jit} at its end
- Arbitrage: $P^{M}_{ji,t+1} = (1+\widetilde{r}_t)\, P^{M}_{jit}$, as for capital
- National accounts record purchases as intermediate consumption

Hulten's Theorem with Input Relationships

Extension: relationships as inherited stocks

$$D_{it} = \sum_{t'=t-\tau}^{t+T} \beta^{g}_{t,t'} \left(S_{C,t'} \frac{\mathrm{d} \log C_{t'}}{\mathrm{d} \log Z_{it}} - S_{L,t'} \frac{\mathrm{d} \log L_{t'}}{\mathrm{d} \log Z_{it}} \right)$$
$$+ \beta^{g}_{t,t+T} \left(S^{+}_{K,t+T} \frac{\mathrm{d} \log K_{t+T+1}}{\mathrm{d} \log Z_{it}} + S^{+}_{M,t+T} \frac{\mathrm{d} \log M_{t+T}}{\mathrm{d} \log Z_{it}} \right), \qquad T \geq 0$$

- The terminal value adds the relationships carried forward
- Efficient production and capital accumulation at supporting prices
- Efficient relationship formation at supporting prices
- Capital and relationships the only inherited stocks, unchanged at $t - \tau$
- Liu and Tsyvinski instead vary the inherited relationships

Environment | Their Experiment | ◂ Extensions | ◂ Capital Accumulation

The Liu–Tsyvinski Experiment

A change in inherited relationships: productivity on its path, capital at t unchanged

$$\sum_{t'=t}^{t+T} \beta^{g}_{t,t'} \left(S_{C,t'} \, \mathrm{d} \log C_{t'} - S_{L,t'} \, \mathrm{d} \log L_{t'} \right) + \beta^{g}_{t,t+T} \left(S^{+}_{K,t+T} \, \mathrm{d} \log K_{t+T+1} + S^{+}_{M,t+T} \, \mathrm{d} \log M_{t+T} \right)$$
$$= \frac{1 + r_{t-1}}{1 + g_{t-1}} \, S^{+}_{M,t-1} \, \mathrm{d} \log M_{t-1}, \qquad T \geq 0$$

- The loss is valued at the compounded value of the relationships inherited at t
- The same efficiency assumptions, at supporting prices

Back

Calibration Approach

Fit: 49 parameters to 380 moments, deviations scaled by the data's standard errors

Calibration Approach

Fit: 49 parameters to 380 moments, deviations scaled by the data's standard errors

Moments		Mainly inform
Covariances: GDP contributions, hours, consumption	275	mechanism; shocks; loadings
Sector TFP growth: variance, AR(1)	22	shock sizes; persistence
5- and 10-year sums of TFP growth	24	trend size and persistence
GDP and hours growth: variance, AR(1)	4	the mechanism
Correlations of the TFP innovations	55	loadings

Covariances: coefficients at $t + h$, $h = 0, \ldots, 4$, on whitened sector TFP growth at t, 1951–2020

Back

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Additional checks: years before $h = 0$ and investment dynamics

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GDP coefficients: implied by the three fitted contributions to GDP growth

Additional checks: years before $h = 0$ and investment dynamics

Comparison economies: no growth, fixed utilization, our shocks, not refitted

- The two geometric economies: $\zeta_j = \delta_j$
- The quantitative model's own benchmark: $\zeta_j = S_{Xj}/S_{Kj}^{+}$ on its path

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Parameters

Estimated mechanism and trend

wealth-effect weight κ	0.23 (0.03)
labor curvature ψ	0, at its bound
adjustment-cost scale	1.47 (0.55)
trend persistence ρ^g	0.76 (0.09)
share learned a year ahead s	1, at its bound

Shocks in three sectors, sizes in pp

	surprise	ρ_i	trend	loading
Construction	0.88	0.24	0.71	0.24
Trade	0.93	−0.43	0.49	0.99
Food & Acc.	1.06	−0.03	0.01	0.48

Fixed: $\beta = 0.975$, log utility, utilization curvature 1.5

All Sectors | What Pins the Mechanism | Back

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- Hours very elastic, though not infinitely: $\overline{C}_t$ multiplies the labor term
- Food & Accommodation: almost all surprise; Trade: surprise mostly common
- Weakly pinned: s has no direct target; ρ^g rests on 70 years of low-frequency moments

All Sectors | What Pins the Mechanism | Back

The Quantitative Model: Capital and Preferences

Capital: depreciation rises with utilization, with r_i^k the steady-state rental rate

$$\delta_i(u) = \delta_{0i} + r_i^k \frac{u^{1.5} - 1}{1.5}$$

- Ψ_i: quadratic in investment growth; zero, with its slope, on the balanced-growth path
- Its curvature: a fixed sector pattern times one estimated scale

Preferences

- $\kappa = 1$: the full wealth effect on hours; $\kappa \to 0$: none
- $\psi = 0$ does not make labor infinitely elastic: $\overline{C}_t$ multiplies the labor term

TFP Process ◂ Back

The Quantitative Model: TFP Process and Measurement

Surprises: one common factor

- Each surprise: one common shock, weighted by the loading, plus a sector shock
- Trade's loading is nonnegative, which fixes the sign of the common shock

Trend: independent across sectors

- Balanced growth from the sectors' TFP trends; first-order solution around it

Measurement: the data's constructions

- Measured TFP is not utilization-adjusted, as in the data

◂ Capital and Preferences ◂ Back

Estimated Shock Parameters: All Sectors

Sizes in pp; standard errors in parentheses

Sector	surprise size (pp)	own persistence ρ_i	trend size (pp)	loading
Mining & Utilities	2.32 (0.27)	0.29 (0.23)	0^b	-0.21 (0.17)
Construction	0.88 (0.22)	0.24 (0.25)	0.71 (0.21)	0.24 (0.14)
Durables	1.58 (0.19)	0.00 (0.14)	0^b	0.77 (0.10)
Nondurables	1.03 (0.10)	0.20 (0.11)	0.01^f	0.44 (0.11)
Trade	0.93 (0.14)	-0.43 (0.08)	0.49 (0.10)	0.99^b
Information	2.16 (0.24)	-0.04 (0.12)	0^b	-0.04 (0.11)
Finance & Insurance	1.93 (0.30)	0.01 (0.11)	0^b	-0.58 (0.11)
Real Estate	1.06 (0.28)	-0.33 (0.30)	0^b	-0.66 (0.21)
Prof. & Business Serv.	1.42 (0.31)	-0.09 (0.37)	0.17 (0.79)	0.23 (0.15)
Educ. & Health	0.15 (3.66)	-0.76 (0.58)	0.56 (0.16)	0.99 (24.78)
Food & Accommodation	1.06 (0.25)	-0.03 (0.24)	0.01 (5.91)	0.48 (0.13)

b: at a bound; f: within 10^{-4} of a bound. Conditional standard errors hold parameters at bounds fixed. Education & Health's surprise size and loading are not determined.

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What Pins the Mechanism

One reset at a time, everything else at the calibration

- Labor curvature $0 \to 0.5$: criterion $514 \to 603$; hours-growth variance $3.1 \to 0.8$
- Wealth effect κ, the brake on hours: hours-growth variance 55 at 0.10, 0.25 at 1
- Adjustment costs $1.47 \to 1$: criterion up 18; GDP-growth variance $6.1 \to 7.5$

Weakly pinned

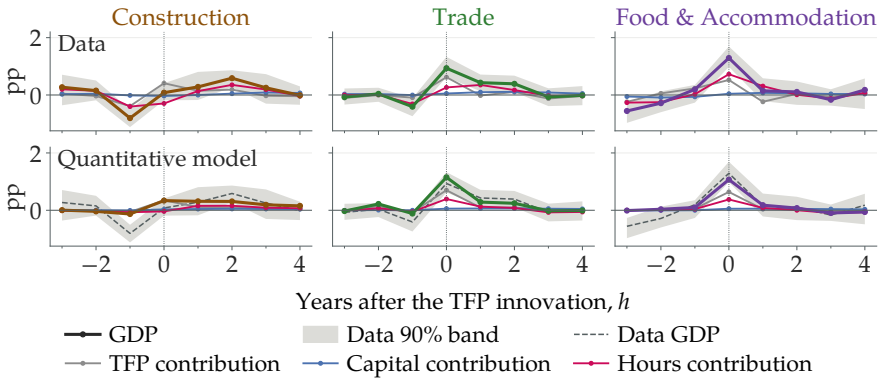
- s, at its bound of 1: no moment targets it directly
- ρ^g and each sector's surprise–trend split: 70 years of low-frequency moments
- The adjustment-cost scale: equally good fits from 1.1 to 1.5

Data: hours-growth variance 6.6, GDP-growth variance 6.3. Local sensitivities, not additive shares.

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GDP Contributions: Data and the Quantitative Model

Regression coefficients: data (top) and the quantitative model (bottom)



- Its persistence comes too little from hours, too much from measured TFP

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The Success Rule, Written before the Fit

Items 1–3 fail; items 4–7 pass

Item, written before the fit	Data	Data 90% band	Quantitative model	Met
1a Construction GDP: year 1 minus year 0	+0.20	[−0.44, +0.81]	−0.03	no
1b Construction GDP: sum of years 1–3	+1.12	[+0.34, +1.81]	+0.81	yes
2a Construction aggregate hours: year 0	−0.49	[−0.77, −0.17]	−0.07	yes
2b Construction aggregate hours: year 2	+0.61	[+0.35, +0.89]	+0.27	no
3 Trade hours contribution: sum of years 1–3	+0.49	[+0.20, +0.83]	+0.13	no
4a Food & Acc. GDP: year 0	+1.29	[+0.98, +1.69]	+1.07	yes
4b Food & Acc. GDP: sum of years 1–3	+0.09	[−0.58, +1.00]	+0.16	yes
5 Trade minus Food & Acc. GDP: sum of years 1–3	+0.66	[−0.06, +1.23]	+0.35	yes
6 Construction: variance of measured TFP growth	+1.99	[+1.29, +2.57]	+1.90	yes
6 Construction: AR(1) of measured TFP growth	+0.44	[+0.28, +0.54]	+0.58	yes
6 Trade: variance of measured TFP growth	+2.28	[+1.67, +2.93]	+2.11	yes
6 Trade: AR(1) of measured TFP growth	−0.01	[−0.20, +0.17]	−0.03	yes
6 Food & Accommodation: variance of measured TFP growth	+1.43	[+0.95, +1.72]	+1.24	yes
6 Food & Accommodation: AR(1) of measured TFP growth	+0.01	[−0.19, +0.16]	−0.01	yes
7a other eight sectors: loss, vs. an earlier fit (423.0)			300.3	yes
7b correlations: loss, vs. earlier mechanism values			67.6	yes

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Fit by Block

Squared standardized deviations

Block	Moments	Loss	Mean
Cross-cov.: TFP contr.	55	44.0	0.80
Cross-cov.: capital contr.	55	111.9	2.03
Cross-cov.: hours contr.	55	81.8	1.49
Cross-cov.: agg. hours	55	79.5	1.45
Cross-cov.: consumption	55	89.5	1.63
TFP variance, AR(1)	22	18.8	0.85
GDP and hours growth	4	10.9	2.72
Low-frequency	24	10.2	0.42
Innovation correlations	55	67.6	1.23
Total	380	514.05	1.35

Aggregate moments

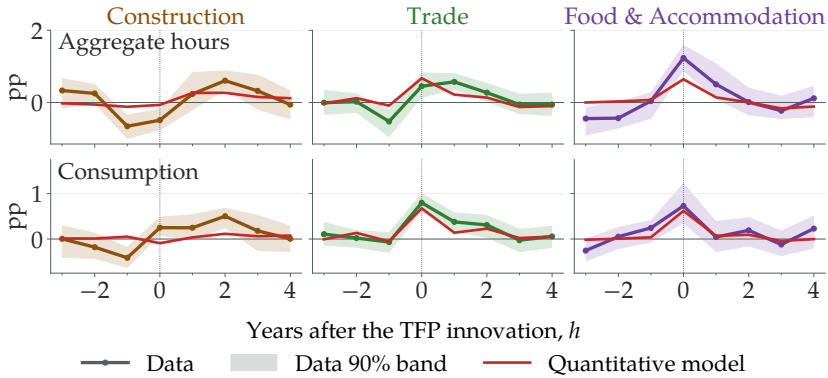
Moment	Data	s.e.	Model
GDP growth: variance	6.262	1.077	6.118
Hours growth: variance	6.620	1.164	3.141
GDP growth: AR(1)	0.137	0.130	0.266
Hours growth: AR(1)	0.194	0.104	0.295

- Worst block per moment: the aggregate moments
- Hours-growth variance: 47% of the data's
- Next: the cross-covariances of the capital contribution

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Hours and Consumption: Three Sectors

Regression coefficients: data and the quantitative model

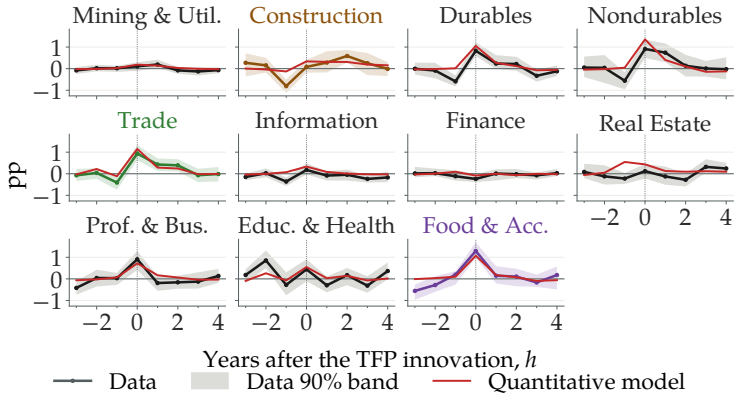


- Construction's hours fall too little on impact and rise too little in year 2

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GDP: All Eleven Sectors

Regression coefficients: data and the quantitative model

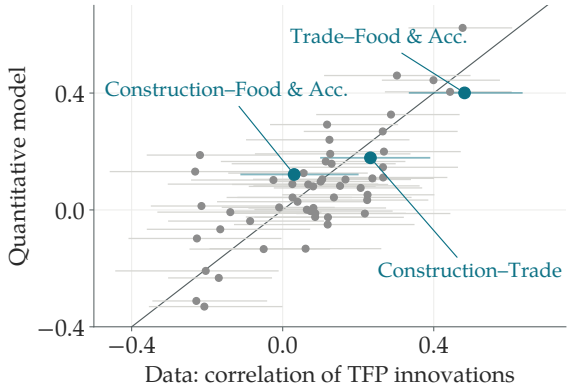


- Largest remaining losses: Information, Mining & Utilities, Real Estate

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Correlations of the TFP Innovations

The 55 sector pairs: data, with its 90% band, and the quantitative model

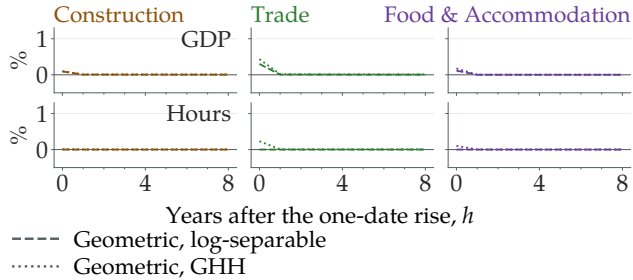


- Correlation across pairs: 0.75; largest miss: Finance and Real Estate

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From the Benchmark to the Quantitative Model

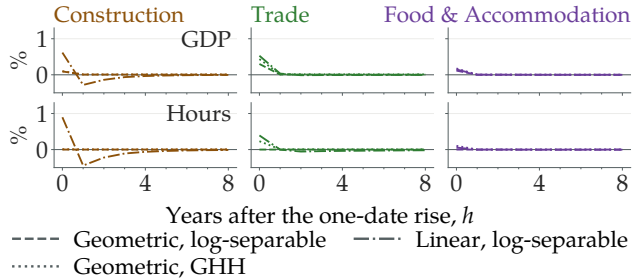
A one-date 1% rise in a sector's productivity, on our eleven-sector production structure



- Geometric, log-separable and Geometric, GHH: small impacts, thin tails

From the Benchmark to the Quantitative Model

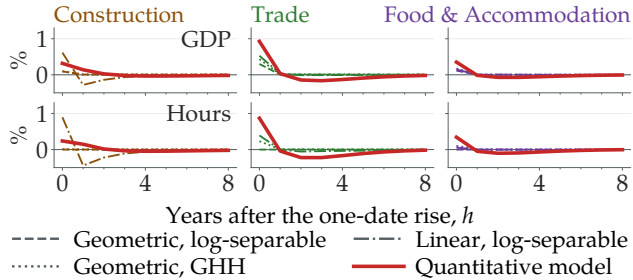
A one-date 1% rise in a sector's productivity, on our eleven-sector production structure



- Geometric, log-separable and Geometric, GHH: small impacts, thin tails
- Linear, log-separable: Construction's GDP jumps, then falls below reference

From the Benchmark to the Quantitative Model

A one-date 1% rise in a sector's productivity, on our eleven-sector production structure



- Geometric, log-separable and Geometric, GHH: small impacts, thin tails
- Linear, log-separable: Construction's GDP jumps, then falls below reference
- Quantitative model: Trade's impact highest; Construction positive at year 1

The Four Economies

Common: our networks and depreciation rates; $\beta = 0.975$; one mobile labor input

- Geometric, log-separable and Geometric, GHH: $\zeta_j = \delta_j$; GHH with unit Frisch
- Linear, log-separable: vom Lehn and Winberry's capital law, no adjustment cost
- Quantitative model: Jaimovich–Rebelo, adjustment costs, utilization, growth

The first three: no growth and fixed utilization

- The same Domar weights and investment shares
- Only accumulation and preferences differ among them
- The quantitative model changes several features at once

vom Lehn and Winberry (2022), Section III: 37 sectors, $\beta = 0.96$; their Section V friction not included

Back

Three Sectors across the Four Economies

One-date 1% productivity rise: responses in percent

	D_i	D_i^x / D_i	GDP, years after the rise				Hours
			0	1	2	3	on impact
Construction							
Geometric, log-separable	0.093	0.89	0.09	0.01	0.01	0.01	0.00
Geometric, GHH	0.093	0.91	0.10	0.01	0.01	0.01	0.01
Linear, log-separable	0.093	8.13	0.62	−0.28	−0.14	−0.06	0.90
Quantitative model	0.107	1.55	0.32	0.14	0.03	−0.02	0.24
Trade							
Geometric, log-separable	0.300	0.21	0.30	0.01	0.01	0.01	0.00
Geometric, GHH	0.300	0.35	0.43	0.02	0.01	0.01	0.23
Linear, log-separable	0.300	1.22	0.53	0.03	−0.01	−0.01	0.39
Quantitative model	0.296	0.80	0.93	0.03	−0.14	−0.16	0.87
Food & Accommodation							
Geometric, log-separable	0.115	0.08	0.12	0.00	0.00	0.00	0.00
Geometric, GHH	0.115	0.24	0.18	0.00	0.00	0.00	0.10
Linear, log-separable	0.115	0.46	0.15	0.00	0.00	0.00	0.06
Quantitative model	0.112	0.65	0.35	−0.02	−0.07	−0.07	0.34

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Geometric and Linear Accumulation

Locally the same: geometric, $\zeta_j = \delta_j$, and linear with the adjustment cost

$$\frac{1}{2}\frac{1-\delta_j}{\delta_j}\left(\frac{X_{jt}}{K_{jt}} - \delta_j\right)^2 K_{jt}$$

- Same reference shares and first-order responses, each law at its own reference
- Curvature $(1-\delta_j)/\delta_j$: from 6.5 to 44.5 across sectors
- So the step to linear accumulation removes a large adjustment cost, locally only

Hours in the linear, log-separable economy

- $\chi L = S_L/S_C$, and spending shares can now move, so hours can respond

Back

GHH: an Exact Common Shift

Geometric, GHH against geometric, log-separable, in every sector i

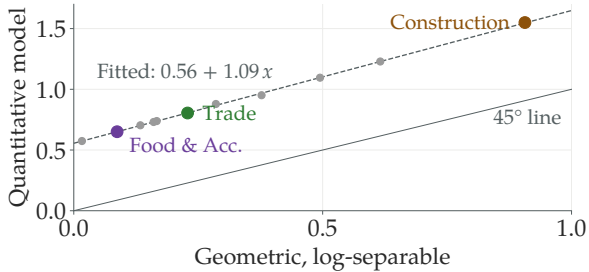
$$\frac{D_i^x(\text{GHH}) - D_i^x(\text{log-sep.})}{D_i - D_i^x(\text{log-sep.})} = \frac{\boldsymbol{S}_X' \boldsymbol{\Omega}' \boldsymbol{\mathcal{L}} \boldsymbol{\gamma}_d (\boldsymbol{I} - \boldsymbol{\alpha}_d)\mathbf{1}}{S_C\left[1 + \psi - \boldsymbol{\theta}' \boldsymbol{\mathcal{L}} \boldsymbol{\gamma}_d (\boldsymbol{I} - \boldsymbol{\alpha}_d)\mathbf{1}\right]} \simeq 0.175$$

- $\boldsymbol{\theta}$: consumption weights; $\psi = 1$, a unit Frisch elasticity
- So $(D_i^x / D_i)_{\text{GHH}} \simeq 0.175 + 0.825\,(D_i^x / D_i)_{\text{log-separable}}$, sector by sector
- Extra hours are proportional to the log-separable impact component
- The invested part of their output enters D_i^x; the Domar weight is unchanged
- The quantitative model's fitted line is a calibration result, not this identity

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Propagation Shares: Quantitative Model and Benchmark

Propagation share of the Domar weight, D_i^x / D_i, sector by sector



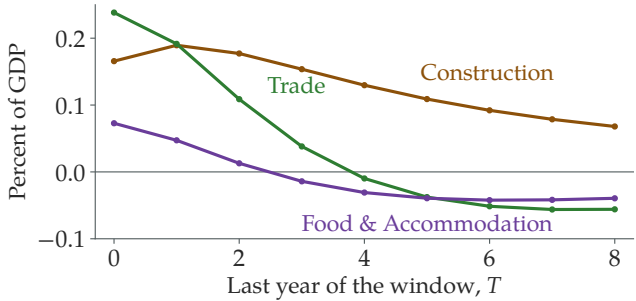
- Same ranking as the Cobb–Douglas benchmark: rank correlation 1
- Fitted line $0.556 + 1.093\,x$, with x the matched benchmark's share
- Benchmark: geometric accumulation, with $\zeta_j = S_{Xj} / S_{Kj}^{+}$ on the model's path

Back

How Long the Gain Stays in Capital

Window ending in year T: D^x_{it} = consumption net of hours + capital carried forward

$$\beta^g_{t,t+T}\, S^+_{K,t+T} \frac{\mathrm{d}\log K_{t+T+1}}{\mathrm{d}\log Z_{it}} \;=\; D^x_{it} - \sum_{t<t'\le t+T} \beta^g_{t,t'} \left(S_{C,t'} \frac{\mathrm{d}\log C_{t'}}{\mathrm{d}\log Z_{it}} - S_{L,t'} \frac{\mathrm{d}\log L_{t'}}{\mathrm{d}\log Z_{it}} \right)$$



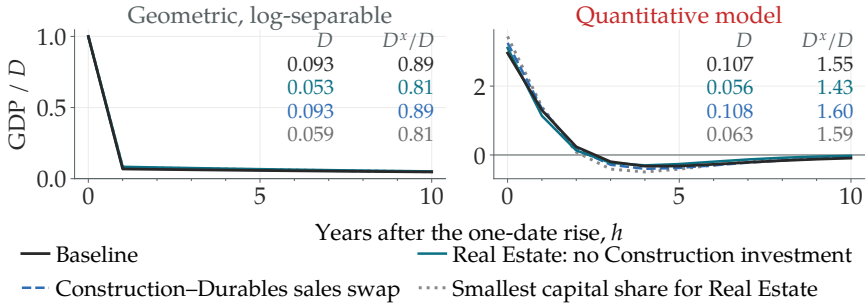
Construction: the value stays in capital for years

Trade, Food & Accommodation: it is delivered within four years

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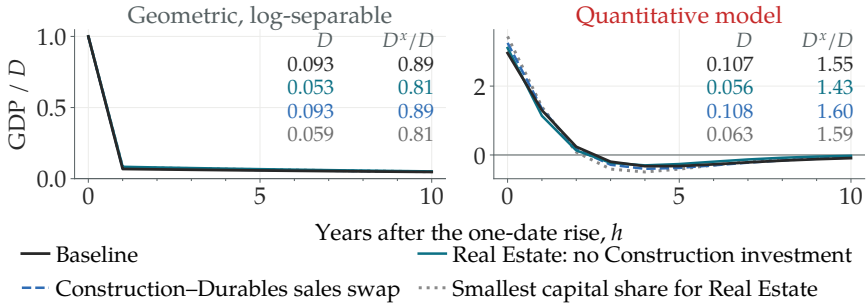
Construction and Real Estate

A one-date 1% rise in Construction's productivity: GDP per unit of its Domar weight



Construction and Real Estate

A one-date 1% rise in Construction's productivity: GDP per unit of its Domar weight

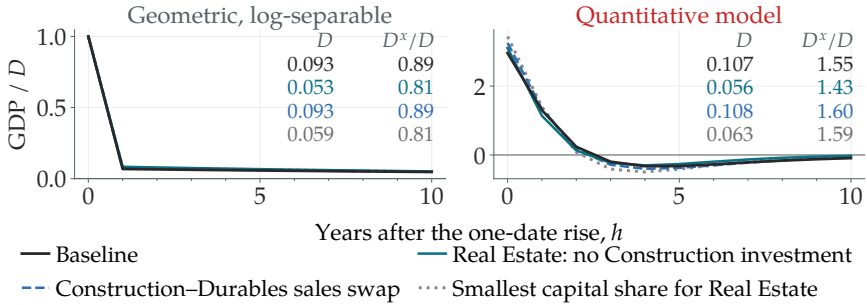


- Remove the direct investment link: Construction's Domar weight falls 43% and 48%
- Its propagation share stays high: D^x/D from 0.89 to 0.81, and from 1.55 to 1.43

The Variants | Linear, Log-Separable | Back

Construction and Real Estate

A one-date 1% rise in Construction's productivity: GDP per unit of its Domar weight



- Remove the direct investment link: Construction's Domar weight falls 43% and 48%
- Its propagation share stays high: D^x/D from 0.89 to 0.81, and from 1.55 to 1.43
- The link makes Construction large and channels its gains into long-lived capital

The Variants | Linear, Log-Separable | Back

Construction and Real Estate: The Variants

Real Estate: 78% of its investment from Construction; capital share 0.94

Three changes, each re-solving the reference equilibrium, nothing refitted

- Real Estate buys no investment from Construction; other supplier shares rescaled
- Construction and Durables swap 0.010 of GDP in sales to Real Estate and Trade
- Real Estate's capital share of value added set to the smallest, 0.14

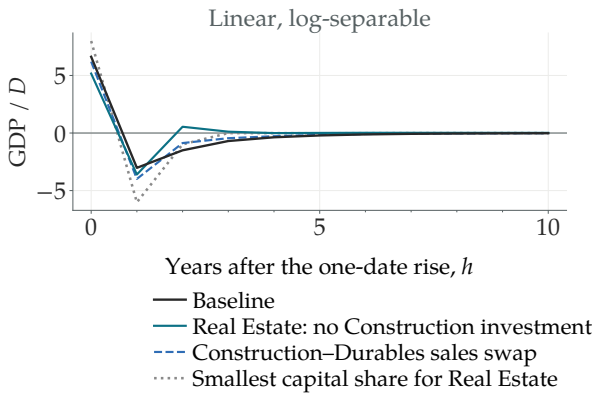
Geometric, log-separable: Real Estate's capital is long-lived, $\delta = 0.022$

- It receives 50% of Construction's propagation component
- It accounts for 31% of Construction's GDP response a year later

Back

Construction and Real Estate: Linear, Log-Separable

A one-date 1% rise in Construction's productivity: GDP per unit of its Domar weight



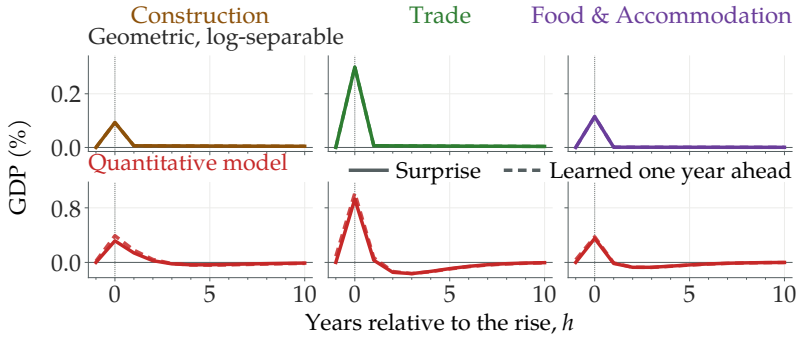
D	D^x/D
0.093	8.13
0.053	5.71
0.093	7.65
0.059	8.78

- The changes alter the path substantially, including the impact and the reversals

Back

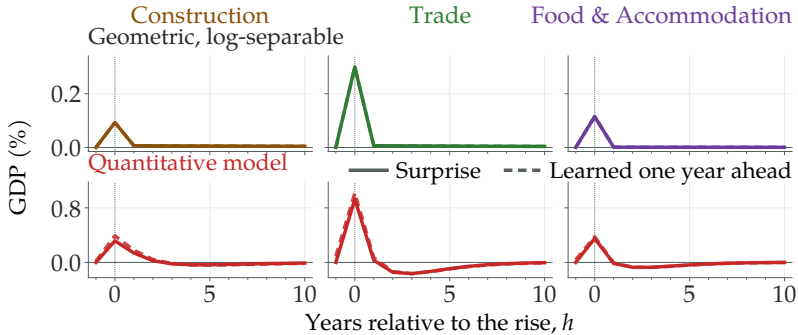
A Surprise and the Same Rise Learned One Year Ahead

A one-date 1% rise, a surprise or learned one year ahead, the longest lead estimated



A Surprise and the Same Rise Learned One Year Ahead

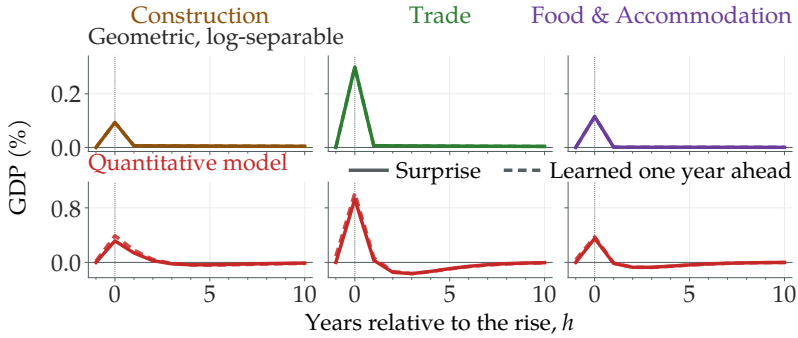
A one-date 1% rise, a surprise or learned one year ahead, the longest lead estimated



- Geometric, log-separable: the lines coincide; nothing moves before the rise

A Surprise and the Same Rise Learned One Year Ahead

A one-date 1% rise, a surprise or learned one year ahead, the longest lead estimated

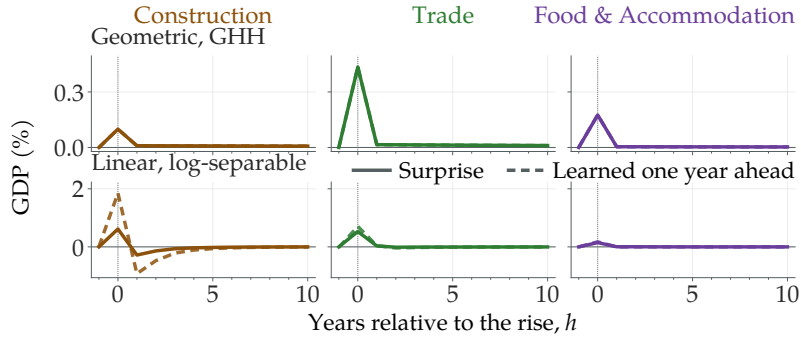


- Geometric, log-separable: the lines coincide; nothing moves before the rise
- Quantitative model: GDP rises before the rise; Construction's impact from 0.32 to 0.39

Other Economies Information Back

A Surprise and News: The Other Economies

A one-date 1% rise in a sector's productivity, learned at the rise or one year earlier

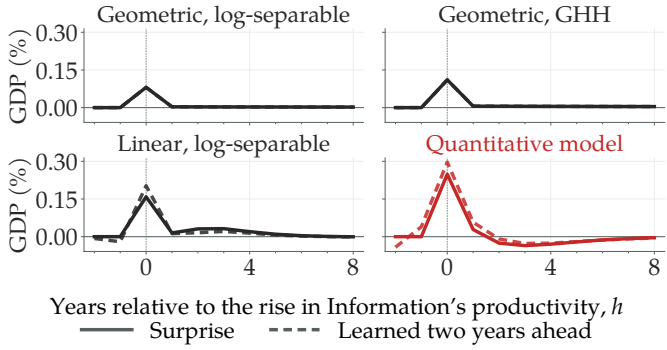


- Geometric, GHH: the lines coincide; nothing moves before the rise
- Linear, log-separable: news triples Construction's impact, 1.85 against 0.62

Back

Information: News Two Years Ahead

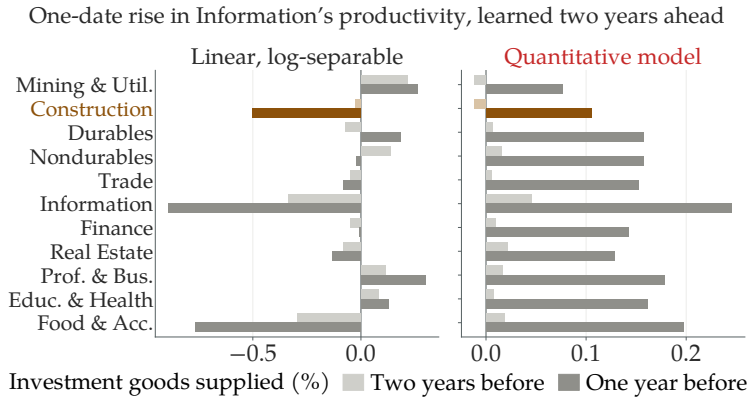
A one-date 1% rise in Information's productivity, a surprise or learned two years ahead



- In the two geometric economies, no real allocation moves before the rise
- Hypothetical: the estimated process has almost no trend news for Information

Investment, One-Date Rise | Investment, Permanent Rise | Back

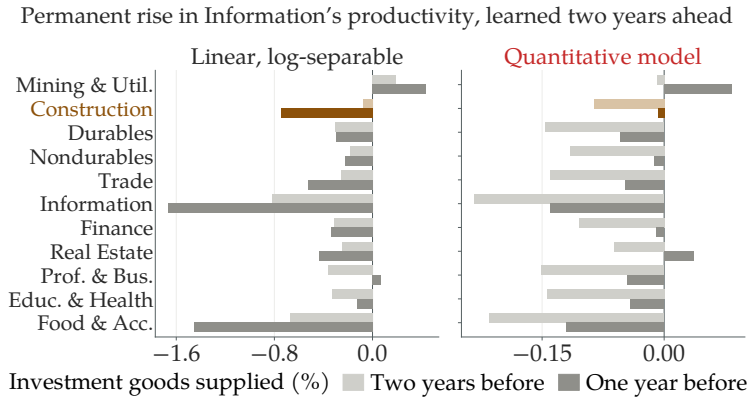
Investment before a One-Date 1% Rise in Information



- Quantitative model: all suppliers expand; Construction, 22% of the increase
- Linear, log-separable: aggregate investment falls before the rise

Permanent Rise Back

Investment before a Permanent 1% Rise in Information

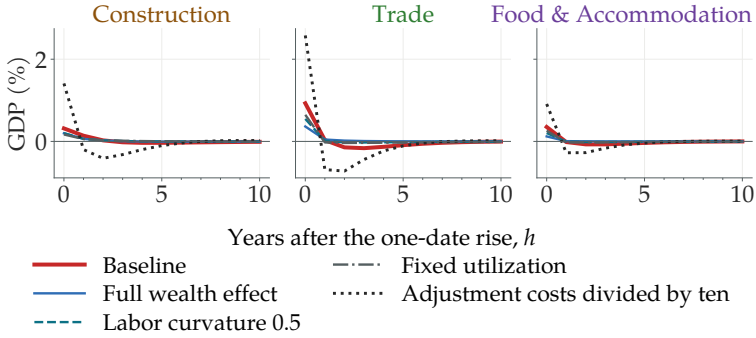


- Quantitative model: aggregate investment falls in the two years before the rise
- Neither experiment produces an investment expansion concentrated in Construction

One-Date Rise Back

Mechanisms within the Quantitative Model

A one-date 1% rise, re-solving after each separate change, nothing refitted



- The first three changes lower Construction's impact from 0.32 to 0.17–0.20
- Adjustment costs divided by ten: impact 1.41, then 0.41 below reference in year two

Back

The Year Before: Data and the Four Economies

Coefficients at $h = -1$: GDP growth on the sector's whitened TFP growth

	Data (90% band)	Quantitative model	Geometric, log-separable	Geometric, GHH	Linear, log-separable
Construction	-0.81 $[-1.12, -0.51]$	-0.13	-0.03	-0.03	0.39
Trade	-0.41 $[-0.74, -0.11]$	-0.11	-0.07	-0.08	-0.21
Durables	-0.59 $[-0.82, -0.32]$	0.02	0.00	0.00	0.10
Information	-0.36 $[-0.57, -0.17]$	0.07	0.00	0.00	0.05

- Quantitative model: negative for Construction and Trade, smaller than in the data
- Not targeted; the estimated mix of surprises and news held fixed

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